SpecEdge: Scalable Edge-Assisted Serving Framework for Interactive LLMs

Jinwoo Park
KAIST
jinwoo520528@kaist.ac.kr

Seunggeun Cho
KAIST
sgn.cho@kaist.ac.kr

Dongsu Han
KAIST
dhan.ee@kaist.ac.kr

Abstract

Large language models (LLMs) power many modern applications, but serving them at scale remains costly and resource-intensive. Current server-centric systems overlook consumer-grade GPUs at the edge. We introduce SpecEdge, an edge-assisted inference framework that splits LLM workloads between edge and server GPUs using a speculative decoding scheme, exchanging only token outputs over the network. SpecEdge employs proactive edge drafting to overlap edge token creation with server verification and pipeline-aware scheduling that interleaves multiple user requests to increase server-side throughput. Experiments show SpecEdge enhances overall cost efficiency by **1.91×** through achieving **2.22×** server throughput, and reduces inter token latency by **11.24%** compared to a server-only baseline, introducing a scalable, cost-effective paradigm for LLM serving. The code is available at `https://github.com/kaist-ina/specedge`

1 Introduction

Large language models (LLMs) have become integral to modern applications such as conversational AI, code generation, and real-time content creation [Dubey et al., 2024, Lozhkov et al., 2024, Achiam et al., 2023, Touvron et al., 2023, Jiang et al., 2023, Brown et al., 2020]. However, scaling LLM deployments to meet growing demand remains challenging when balancing operational costs against latency requirements.

A compelling opportunity exists to dramatically reduce LLM serving costs by leveraging consumer-grade GPUs at the network edge. The GeForce RTX 4090 [Nvidia, 2025a] delivers up to 330.3 TFLOPS for FP16 tensor operations with FP16 accumulate [Nvidia, 2024a], exceeding the 312 TFLOPS of the data-center-class A100 [Nvidia, 2025b], at 14.43x lower cost [GCP, 2025, Vas, 2025]. With the widespread availability of these powerful edge devices [Valve, 2024, Nvidia, 2024b], an edge-assisted inference approach that offloads computation to these cost-effective resources could fundamentally transform the economics of LLM deployment.

Despite this opportunity, existing inference architectures fail to leverage these edge resources effectively. Current parallelization techniques [Shoeybi et al., 2019, Rasley et al., 2020, Aminabadi et al., 2022] that split computation within data centers break down over public internet conditions, where high latency and limited bandwidth make frequent communication of intermediate results impractical. Mainstream approaches like tensor and pipeline parallelism rely on high-bandwidth, low-latency interconnects such as NVLink or InfiniBand [Nvidia, 2025c,d]. In wide-area networks (WANs), transferring intermediate model states between edge and server GPUs quickly becomes prohibitive, preventing meaningful collaboration between these heterogeneous resources.

In this paper, we present SpecEdge, the first practical edge-assisted inference framework that fundamentally reduces LLM serving costs by splitting computation between consumer-grade edge GPUs and cloud servers. The core innovation of SpecEdge is its ability to effectively coordinate edge and server resources over wide-area networks—a capability previously unattainable with traditional

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SpecEdge：用于交互式大语言模型的可扩展边缘辅助服务框架

Jinwoo Park
KAIST
jinwoo520528@kaist.ac.kr

Seunggeun Cho
KAIST
sgn.cho@kaist.ac.kr

Dongsu Han
KAIST
dhan.ee@kaist.ac.kr

摘要

大语言模型（ LLMs ）驱动了许多现代应用，但大规模部署它们仍然成本高且资源密集。现有以服务器为中心的系统忽略了边缘的消费级 GPU。我们提出了 SpecEdge，一种边缘辅助推理框架，它使用推测解码方案将 LLM 工作负载在边缘和服务器 GPU 之间分配，仅通过网络交换 token 输出。SpecEdge 采用主动式边缘草稿生成，将边缘 token 生成与服务器验证重叠，并通过流水线感知调度交错多个用户请求以提高服务器端吞吐量。实验表明，SpecEdge 通过实现服务器吞吐量提高 **2.22×**，整体成本效率提高 **1.91×**，相比仅服务器基线，token 间时延降低 **11.24%**，为 LLM 服务引入了一种可扩展、成本效益高的新范式。代码可在 https://github.com/kaist-ina/specedge 获取。

1 引言

大语言模型（ LLMs ）已经成为现代应用程序的核心部分，如对话式人工智能、代码生成和实时内容创作 [Dubey 等 , 2024, Lozhkov 等 , 2024, Achiam 等 , 2023, Touvron 等 , 2023, Jiang 等 , 2023, Brown 等 , 2020]。然而，在平衡运营成本与时延要求时，扩大 LLM 部署以满足不断增长的需求仍然具有挑战性。

通过在网络边缘利用消费级 GPU，有一个极具吸引力的机会可以显著降低大语言模型的服务成本。GeForce RTX 4090 [Nvidia, 2025a] 在 FP16 张量操作和 FP16 累加运算中可提供高达 330.3 TFLOPS 的性能 [Nvidia, 2024a], , 超过了数据中心级 A100 [Nvidia, 2025b], 的 312 TFLOPS 性能，其成本低 14.43 倍 [GCP, 2025, Vas, 2025]。随着这些强大边缘设备的广泛可用 [Valve, 2024, Nvidia, 2024b], , 一种将计算负载转移到这些性价比高的资源上的边缘辅助推理方法可能从根本上改变大语言模型部署的经济性。

尽管有这个机会，现有的推理架构未能有效利用这些边缘资源。现有的并行化技术 [Shoeybi 等人，2019，Rasley 等人，2020，Aminabadi 等人，2022] 在数据中心内部拆分计算的方法，在公共互联网环境下会失效，因为高时延和有限的带宽使得频繁传输中间结果变得不切实际。主流方法如张量并行和流水线并行依赖于高带宽、低时延的互连，如 NVLink 或 InfiniBand [Nvidia，2025c,d]。在广域网（ WAN ）中，在边缘与服务器 GPU 之间传输中间模型状态会迅速变得不可行，从而阻止这些异构资源之间进行有意义的协作。

在本文中，我们提出了 SpecEdge，这是第一个实际的边缘辅助推理框架，通过在消费级边缘 GPU 和云服务器之间分配计算，从根本上降低了大语言模型（ LLM ）的服务成本。SpecEdge 的核心创新在于其能够有效地在广域网中协调边缘和服务器资源 —— 这是传统方法以前无法实现的能力。

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parallelization techniques. To achieve this, SpecEdge adopts a speculative decoding paradigm that divides LLM inference into *edge drafting* and *server verification*, exchanging only token outputs rather than full model states. This approach dramatically reduces bandwidth requirements while minimizing communication rounds.

To make this edge-assisted paradigm practical in real-world deployment scenarios, SpecEdge implements two key enabling techniques. Our *Proactive Edge Drafting* allows edge GPUs to continue generating tokens while awaiting server verification, effectively masking network and verification latency with local computation. Complementing this, our *Server-side Pipeline-aware Scheduling* orchestrates verification requests from multiple users through intelligent batching to maintain high GPU utilization. Together, these techniques ensure that the inherent cost advantages of edge-assisted inference are not undermined by network constraints or inefficient resource utilization.

Our evaluation validates the effectiveness of this edge-assisted approach, demonstrating that SpecEdge achieves **1.91×** better cost efficiency while increasing server-side throughput by **2.22×** and reducing inter token latency by **11.24%**. These improvements persist even under challenging wide-area network conditions, outperforming server-only baselines with zero network delays. By effectively harnessing widely available edge GPUs, SpecEdge establishes a new paradigm for scalable and cost-effective LLM serving that addresses the growing computational demands of generative AI applications.

2 Background and Related Work

LLM serving systems. Modern LLM frameworks address latency, throughput, and resource efficiency challenges through various optimizations. DeepSpeed-Inference [Rasley et al., 2020] and TensorRT-LLM [Nvidia, 2024c] leverage low-level GPU optimizations, model parallelism, and quantization, while vLLM [Kwon et al., 2023] introduces PagedAttention for efficient memory management. Modern parallelization strategies [Shoeybi et al., 2019, Aminabadi et al., 2022] enhance the performance with multiple GPUs. However, these approaches depend on data-center GPUs connected via specialized interconnects (InfiniBand, NVLink) with throughput exceeding hundreds of GB/s [Nvidia, 2025c]—speeds unattainable over wide area networks (WANs).

Speculative decoding. Speculative decoding [Leviathan et al., 2023, Chen et al., 2023] reduces latency by having a smaller auxiliary model generate multiple candidate tokens for parallel verification by the main model. The process involves three phases: drafting candidates, verification, and reconciliation for generated tokens. Recent advances have focused on more efficient drafting methods, either using lighter auxiliary models or the target model with reduced parameters [Bhendawade et al., 2024, Cai et al., 2024, Li et al., 2024, Stewart et al., 2024, Zhang et al., 2023].

Tree-based speculative decoding. Standard speculative decoding suffers from exponentially declining acceptance rates as sequence length increases [Leviathan et al., 2023, Chen et al., 2023]. Tree-based approaches [Miao et al., 2024, Chen et al., 2024, Svirschevski et al., 2024, Cai et al., 2024, Sun et al., 2024] address this by exploring multiple paths simultaneously. Sequoia [Chen et al., 2024] and SpecExec [Svirschevski et al., 2024] further optimize by pruning unpromising branches.

Distributed LLM serving. Split-inference approaches like Petals [Borzunov et al., 2024] and Helix [Mei et al., 2024] distribute model layers and pipeline requests across multiple devices, improving throughput over memory offloading methods [Ren et al., 2021, Pudipeddi et al., 2020], but introduce network delays that increase latency compared to data center solutions. Several works [Timor et al., 2025, Liu et al., 2025, McDanel, 2024] utilize multiple GPU devices within a server node to overlap draft and verification tasks of speculative decoding for faster inference, yet with an exchange of higher cost as they require additional server devices per query.

On-device LLM inference. Frameworks like MLC LLM [MLC team, 2023-2024] and Web LLM enable fully on-device inference with smaller or quantized models. While recent approaches [Svirschevski et al., 2024, Song et al., 2024, Xue et al., 2024] showcase the potential of leveraging user devices, they compromise output quality and latency. In contrast, our approach retains the high-quality verification stage on the server—achieving a hybrid solution that outperforms purely centralized or fully local inference.

并行化技术。为实现这一点，SpecEdge 采用了一种推测解码范式，将 LLM 推理分为边缘草稿生成和服务器验证，仅交换 token 输出而不是完整的模型状态。这种方法显著减少了带宽需求，同时最小化了通信轮次。

为了使这种边缘辅助范式在现实部署场景中切实可行，SpecEdge 实施了两项关键的启用技术。我们的 主动式边缘草稿生成 允许边缘 GPU 在等待服务器验证的同时继续生成 token，有效地通过本地计算掩盖网络和验证时延。与此互补，我们的 服务器端流水线感知调度通过智能批处理协调来自多个用户的验证请求，以保持 GPU 的高利用率。这些技术共同确保了边缘辅助推理固有的成本优势不会被网络限制或资源利用效率低下所削弱。

我们的评估验证了这种边缘辅助方法的有效性，显示 SpecEdge 实现了 **1.91×** 更高的成本效率，同时将服务器端吞吐量提高了 **2.22×**，并将 token 间时延降低了 **11.24%**。即使在具有挑战性的广域网条件下，这些改进仍然存在，性能优于零网络时延的仅服务器基线。通过有效利用广泛可用的边缘 GPU，SpecEdge 为可扩展且经济高效的 LLM 服务确立了一种新的范式，以应对生成式 AI 应用日益增长的计算需求。

2 背景与相关工作

LLM 服务系统。现代 LLM 框架通过各种优化解决时延、吞吐量和资源效率的挑战。DeepSpeed-Inference [Rasley 等人，2020] 以及 TensorRT-LLM [Nvidia, 2024c] 利用低级 GPU 优化、模型并行和量化，而 vLLM [Kwon 等人，2023] 引入了 PagedAttention 以实现高效的内存管理。现代并行化策略 [Shoeybi 等人，2019，Aminabadi 等人，2022] 通过多 GPU 提升性能。然而，这些方法依赖于通过专用互连（ In niBand 、 NVLink ）连接的数据中心 GPU，其吞吐量超过数百 GB/s [Nvidia, 2025c]—— 这种速度在广域网（ WAN ）上无法实现。

推测解码。推测解码 [Leviathan 等，2023，Chen 等，2023] 通过让一个较小的辅助模型生成多个候选 token，以供主模型并行验证，从而降低时延。该过程包括三个阶段：候选 token 的草稿生成、验证以及生成 token 的调和。最近的进展集中在更高效的草稿生成方法，要么使用更轻量的辅助模型，要么使用参数减少的目标模型 [Bhendawade 等，2024，Cai 等，2024，Li 等，2024，Stewart 等，2024，Zhang 等，2023]。

基于树的推测解码。 标准推测解码在序列长度增加时会遭遇指数下降的接受率 [Leviathan 等 , 2023, Chen 等 , 2023]. 基于树的方法 [Miao 等 , 2024, Chen 等 , 2024, Svirschevski 等 , 2024, Cai 等 , 2024, Sun 等 , 2024] 通过同时探索多条路径来解决这一问题。Sequoia [Chen 等 , 2024] 以及 SpecExec [Svirschevski 等 , 2024] 通过剪枝不太可能成功的分支进一步优化。

分布式大语言模型服务。 像 Petals 这样的分拆推理方法 [Borzunov 等， 2024] 以及 Helix [Mei 等, 2024] 在多个设备上分布模型层和流水线请求，相对于内存卸载方法提高吞吐量 [Ren 等，2021, Pudipeddi 等， 2020], 但会引入网络时延，使得时延相比数据中心解决方案增加。若干工作 [Timor 等，2025，Liu 等，2025，McDanel， 2024] 在服务器节点内使用多个 GPU 设备来重叠推测解码的草稿和验证任务，从而实现更快的推理，但代价是更高，因为每个查询需要额外的服务器设备。

设备端大语言模型推理。 像 MLC LLM 这样的框架 [MLC 团队，2023-2024] 以及 Web LLM 实现了使用较小或量化模型的完全设备端推理。虽然近期的方法 [Svirschevski 等，2024，Song 等，2024，Xue 等，2024] 展示了利用用户设备的潜力，但它们会影响输出质量和时延。相比之下，我们的方法在服务器端保留了高质量的验证阶段 —— 实现了一种混合解决方案，其表现优于纯集中式或完全本地的推理。

3 Problem and Motivation

Serving large language models (LLMs) presents significant computational challenges due to their resource-intensive nature. While data centers rely on expensive H100 and A100 GPUs, abundant computing resources exist at the edge in the form of consumer-grade GPUs. As Figure 1 demonstrates, edge devices like the RTX 4090 and RTX 3090 generate tokens at approximately **30-50×** lower cost than server-class GPUs when running small but capable language models like Qwen2-0.5B. Despite this cost advantage and widespread availability, current LLM serving architectures fail to incorporate these edge resources, creating a substantial missed opportunity for distributed inference that could reduce costs while maintaining high-quality LLM service.

Conventional split computing. One approach to utilizing edge GPUs is split computation across edge and server resources. Split computing has been extensively studied in machine learning literature [Kang et al., 2017, Zhou et al., 2019, Wang et al., 2020], predating transformer architectures. Traditionally, this approach divides neural network layers between server and edge, reducing server computational load or minimizing network communication by transmitting intermediate tensors rather than raw input data. However, this method is poorly suited for LLM serving due to the unique characteristics of transformers [Vaswani, 2017]:

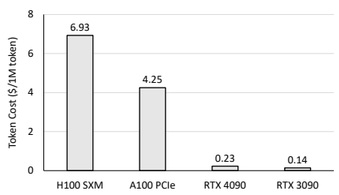


Figure 1: Token generation cost comparison with Qwen2-0.5B model.

Limitation #1: Excessive latency. While layer-wise splitting has been used in distributed peer-to-peer systems [Borzunov et al., 2024, Mei et al., 2024] to enable serving large models otherwise infeasible on consumer GPUs, these approaches prioritize feasibility over performance. This is because, unlike earlier ML applications [He et al., 2016, Redmon, 2016, Diwan et al., 2023] that typically involve only a few communication rounds, LLM applications require communication for every generated token due to their autoregressive nature. In our distributed network setting (§5.3), layer-wise splitting increases latency by 2.35x compared to the server-only solution.

Limitation #2: Ineffectiveness in reducing I/O. LLM inference is inherently constrained by memory I/O [Pope et al., 2023], as generating a single token requires accessing the entire model's parameters. Transformer architectures, with operations like general matrix-vector multiplication (GEMV), have low arithmetic intensity relative to their memory I/O demands. While layer-wise split computing reduces computational load by offloading layers to the edge, it fails to alleviate the I/O-bound nature of inference. A more effective approach is batch-verifying multiple candidate tokens, which significantly reduces the I/O overhead per token and increases GPU throughput.

4 SpecEdge Design

Our goal is to combine consumer-grade edge GPUs with server resources for cost-efficient LLM serving without compromising quality or latency.

4.1 Disaggregated LLM Decoding

To overcome conventional split computing limitations, we offload token drafting to edge GPUs while keeping verification on servers. Unlike traditional LLM serving, where servers handle the entire process, SpecEdge disaggregates speculative decoding into two distinct phases: edge-based *drafting* and server-based *verification*.

In our novel design, edge GPUs generate candidate tokens and send them to the server, which verifies them in a single forward pass. The server returns both the verified tokens and one additional token to the edge device, which then updates its sequence, KV cache, and continues drafting. This cycle repeats until an end-of-sequence token is generated. Critically, this approach preserves the exact output distribution of the server model [Leviathan et al., 2023]—even when using different models for drafting and verification, the final output is guaranteed to be sampled from the same distribution as if generated by the server model alone.

3 问题与动机

为大语言模型（ LLMs ）提供服务由于其资源密集型特性而面临显著的计算挑战。虽然数据中心依赖昂贵的 H100 和 A100 GPU，但边缘存在丰富的计算资源，以消费级 GPU 的形式存在。如图 1 所示，边缘设备如 RTX 4090 和 RTX 3090 在运行像 Qwen2-0.5B 这样的小型但功能强大的语言模型时，每生成一个 token 的成本比服务器级 GPU 低大约 **30-50×** 。尽管具有这种成本优势且广泛可用，但当前的 LLM 服务架构未能整合这些边缘资源，从而错过了通过分布式推理在降低成本的同时保持高质量 LLM 服务的重大机会。

传统分割计算。一种利用边缘 GPU 的方法是在边缘和服务器资源之间进行分割计算。分割计算在机器学习文献中已经被广泛研究 [Kang 等 , 2017, Zhou 等 , 2019, Wang 等 , 2020]，早于变换器架构。传统上，这种方法将神经网络层在服务器和边缘之间划分，通过传输中间张量而非原始输入数据来降低服务器的计算负载或最小化网络通信。然而，由于变换器的独特特性，这种方法并不适合 LLM 服务 [Vaswani, 2017]:

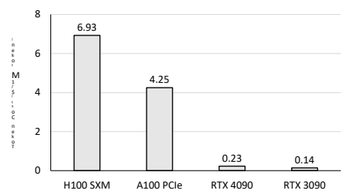


图 1：与 Qwen2-0.5B 模型的 token 生成成本比较。

限制 #1：过高的时延。虽然层级拆分已经在分布式点对点系统中使用 [Borzunov 等 , 2024, Mei 等 , 2024] 以支持在普通 GPU 上无法实现的大模型，这些方法更注重可行性而非性能。这是因为，与早期的机器学习应用不同 [He 等 , 2016, Redmon, 2016, Diwan 等 , 2023] 早期应用通常只涉及少数通信轮次，而由于自回归特性，LLM 应用每生成一个 token 就需要通信。在我们的分布式网络设置中（ §5.3 ），相较于仅服务器解决方案，层级拆分将时延增加了 2.35 倍。

限制 #2：在减少 I/O 方面效果有限。LLM 推理本质上受制于内存 I/O。[Pope 等人，2023]，指出，生成单个 token 需要访问整个模型的参数。Transformer 架构中，例如通用矩阵 - 向量乘法（ GEMV ）等操作，相对于其内存 I/O 需求算力强度较低。虽然逐层拆分计算通过将层卸载到边缘设备来减轻计算负载，但无法缓解推理的 I/O 受限特性。一个更有效的方法是批量验证多个候选 token，这显著降低每个 token 的 I/O 开销并提高 GPU 吞吐量。

4 SpecEdge 设计

我们的目标是将消费级边缘 GPU 与服务器资源结合起来，在不影响质量或时延的情况下实现成本高效的大语言模型服务。

4.1 分解型大语言模型解码

为了克服传统分割计算的限制，我们将 token 草稿生成移到边缘 GPU 上，同时在服务器上保持验证。与传统的大语言模型服务不同，传统服务中服务器处理整个过程，SpecEdge 将推测解码拆分为两个不同的阶段：基于边缘的草稿生成和基于服务器的验证。

在我们新的设计中，边缘 GPU 生成候选 token 并将其发送到服务器，服务器在一次前向传递中验证这些 token。服务器返回经过验证的 token 以及一个额外的 token 给边缘设备，边缘设备随后更新其序列、 KV 缓存，并继续生成。这一循环重复进行，直到生成序列结束 token。关键是，这种方法保持了服务器模型的精确输出分布 [Leviathan 等，2023]—— 即使在使用不同模型进行生成和验证时，最终输出也保证从与仅由服务器模型生成时相同的分布中采样。

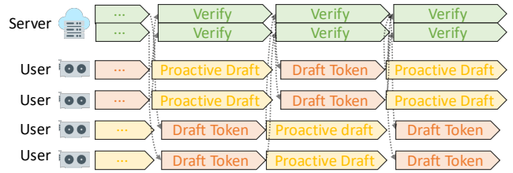


Figure 2: Abstract timeline of SpecEdge with *draft* (edge-side) and *verify* (server-side) inference concept.

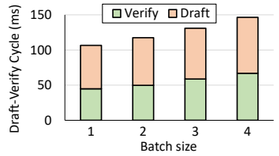


Figure 3: A speculative decoding cycle composed of draft and verify stages.

As illustrated in Figure 2, this disaggregated approach enables the server to focus solely on verification while edge devices handle the autoregressive drafting process. By shifting much of the drafting time to inexpensive edge GPUs, we significantly reduce the overall serving cost while increasing server throughput.

Figure 3 quantifies the potential advantage of this approach by showing that draft stages account for the majority of the draft-verify cycle across different batch sizes. By offloading this dominant component to edge devices, SpecEdge can substantially reduce server time and improve overall system efficiency.

This disaggregated approach promises two fundamental advantages over conventional split computing:

- **Resolving excessive latency (Limitation #1):** Unlike layer-wise splitting that requires per-token communication rounds, SpecEdge dramatically reduces network interaction by a speculative decoding approach. This minimizes both the number of client-server round trips and the volume of data transferred, eliminating the excessive latency inherent in traditional split approaches.
- **Addressing I/O bottlenecks (Limitation #2):** By enabling batch verification of multiple tokens in a single server-side forward pass, SpecEdge amortizes the cost of model parameter access across multiple token verifications. This increases the arithmetic intensity of each operation, directly addressing the I/O-bound nature of LLM inference and substantially improving GPU throughput.

However, to fully realize these benefits, we must overcome two critical challenges that emerge when separating drafting and verification across different devices:

- **Potential latency increase:** In conventional speculative decoding, drafting and verification happen sequentially on the same device with minimal transition overhead. In our disaggregated setting, naïvely implementing this sequence would add network round-trip delays to each draft-verify cycle, potentially negating our latency advantages and deteriorating user experience.
- **Risk of server underutilization:** Without careful coordination, server GPUs would remain idle while waiting for edge devices to complete drafting. This inefficiency could severely limit throughput and undermine the cost benefits of our approach, particularly when serving multiple concurrent users with varying workload patterns.

To unlock the full potential of disaggregated speculative decoding, SpecEdge introduces two key innovations: Proactive edge drafting (§4.2), which masks network latency by continuously generating token candidates without waiting for verification results, and pipeline-aware scheduling (§4.3), which maximizes server GPU utilization by intelligently batching verification requests from multiple users. Together, these techniques enable SpecEdge to achieve both low latency and high cost efficiency.

4.2 Proactive Draft Generation at the Edge

Our key insight is to eliminate the idle time by continuing draft generation during server verification. The edge GPU performs two types of drafting: initial drafting to generate the first batch of candidate tokens, and proactive drafting that continues during server verification. When the edge GPU sends n candidate tokens to the server, it immediately continues drafting additional tokens without waiting for verification results. If any token is rejected during verification, these proactively generated tokens are discarded. However, when all n tokens are accepted and the server's bonus token matches the first proactively drafted token—a scenario we call "complete draft alignment"—these additional tokens can be immediately utilized, effectively hiding network and verification latency. This approach significantly reduces end-to-end latency by overlapping computation with communication, eliminating waiting periods between draft-verify cycles. Simultaneously, it improves server throughput by

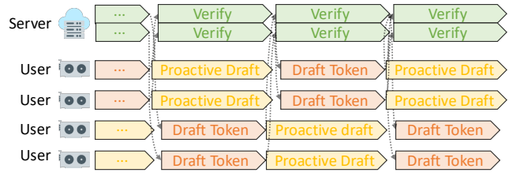


图 2：SpecEdge 的抽象时间线，包含草稿（边缘侧）和验证（服务器端）推理概念。

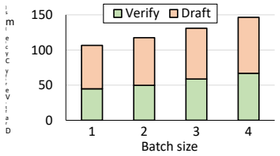


图 3：由草稿和验证阶段组成的推测解码循环。

如图 2 所示，这种拆分方法使服务器能够专注于验证，而边缘设备则处理自回归草稿生成过程。通过将大部分草稿生成时间转移到廉价的边缘 GPU，我们显著降低了整体服务成本，同时提高了服务器吞吐量。

图 3 量化了这种方法的潜在优势，显示了在不同批处理大小下，草稿阶段占据了草稿 - 验证循环的主要部分。通过将这一主要环节卸载到边缘设备，SpecEdge 可以显著减少服务器时间并提高整体系统效率。

This 分解方法相比传统的分布式计算承诺有两个基本优势 g:

- **解决过高时延问题（限制 #1）：**与需要逐 token 通信轮次的分层拆分不同，SpecEdge 通过一种推测解码方法大幅减少了网络交互。这既最小化了客户端与服务器的往返次数，也减少了传输数据的量，从而消除了传统拆分方法固有的过高时延。
- **解决 I/O 瓶颈（限制 #2）：**通过在单次服务器端前向传递中启用对多个 token 的批量验证，SpecEdge 在多个 token 验证之间分摊了模型参数访问的开销。这提高了每次操作的算术强度，直接应对了大语言模型推理的 I/O 受限特性，并显著提升了 GPU 吞吐量。

然而，要充分实现这些好处，我们必须克服在不同设备上分离草稿生成和验证时出现的两个关键挑战：

- **潜在时延增加：**在传统的推测解码中，草稿和验证在同一设备上顺序进行，转换开销最小。在我们的分离式环境中，天真地实现这一序列会在每个草稿 - 验证周期增加网络往返时延，可能抵消我们的时延优势并恶化用户体验。
- **服务器未充分利用的风险：**如果没有仔细协调，当边缘设备完成草稿制作时，服务器 GPU 将保持空闲。这种低效率可能会严重限制吞吐量，并削弱我们方法的成本效益，尤其是在为多个具有不同工作负载模式的并发用户服务时。

为了释放分解式推测解码的全部潜力，SpecEdge 引入了两项关键创新：主动边缘草稿生成 (§4.2)，通过不断生成 token 候选而不等待验证结果来掩盖网络时延，以及流水线感知调度 (§4.3)，通过智能地批处理来自多个用户的验证请求来最大化服务器 GPU 利用率。这些技术结合起来，使 SpecEdge 能够同时实现低时延和高成本效率。

4.2 边缘的主动草稿生成

我们的关键洞见是在服务器验证期间继续生成草稿，从而消除空闲时间。边缘 GPU 执行两类草稿生成：初始草稿生成用于产生第一批候选 token，主动式草稿生成则在服务器验证期间继续进行。当边缘 GPU 向服务器发送 n 个候选 token 后，它会立即继续生成额外 token，而不等待验证结果。若验证过程中有任何 token 被拒绝，这些主动生成的 token 就会被丢弃。不过，当全部 n 个 token 均被接受，且服务器的奖励 token 与主动生成的第一个 token 一致时，我们将这种情况称为 “ 完全草稿对齐 ”。此时，额外生成的 token 可立即使用，从而有效掩蔽网络和验证时延。通过使计算与通信重叠，这种方法显著降低了端到端时延，消除了草稿生成与验证周期之间的等待。同时，它还通过以下方式提高服务器吞吐量：

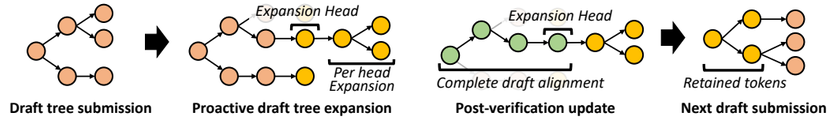


Figure 4: Illustrative example of a proactive draft tree expansion.

Figure 5: Post-verification update with complete draft alignment and subsequent draft submission.

ensuring a continuous flow of verification requests, keeping expensive server GPUs highly utilized rather than idle while waiting for edge devices to complete drafting.

Initial drafting. SpecEdge employs state-of-the-art tree-based drafting [Svirschevski et al., 2024]. However, SpecEdge is designed to be future-proof and not tied to any specific drafting technique. It simply requires a draft phase that produces candidate tokens and is compatible with various speculative decoding approaches, including tree-based methods [Chen et al., 2024, Miao et al., 2024], lossy/lossless methods, and earlier single-sequence schemes [Leviathan et al., 2023, Chen et al., 2023]. As the field evolves, any advances in speculative decoding can be seamlessly integrated.

Proactive draft tree expansion. Unlike conventional speculative decoding that stops after initial token generation, our approach continues drafting proactively to anticipate future server responses. The expected gain from this proactive expansion can be expressed as:

$$\mathbb{E}(\text{Gain}) = P_{\text{align}} \cdot P_{\text{match}\,|\,\text{align}} \cdot \left(\frac{T_{\text{draft}}}{H_{\text{expan}}} - 1 \right) \tag{1}$$

where P_{align} represents the probability of alignment between verified and drafted sequences, $P_{\text{match}\,|\,\text{align}}$ is the probability that the server's extra token matches an expansion head given alignment, T_{draft} is the total number of proactively drafted tokens, and H_{expan} is the number of expansion heads (leaf nodes from which drafting continues).

This formulation reveals a fundamental trade-off: increasing H_{expan} improves alignment probability (P_{align}) but decreases the token preservation ratio ($\frac{T_{\text{draft}}}{H_{\text{expan}}} - 1$), while fewer heads reduce alignment probability but significantly increase preservation when alignment succeeds. Naïvely applying the initial drafting strategy—treating every leaf node as an expansion head—yields negligible gains despite maximizing P_{align}. Our empirical results demonstrate a counterintuitive but optimal approach: after generating the initial draft tree, SpecEdge identifies the single path with the highest cumulative log probability and continues drafting exclusively from that node (Figure 4). This focused strategy maximizes expected gain by producing significantly higher returns when alignment occurs, despite the reduced alignment probability.

Post-verification update. When server verification completes, the edge GPU receives both the accepted tokens and one additional token generated during the server's forward pass. At this point, SpecEdge compares the server's output with its proactively drafted tokens. Also, the edge updates its draft model KV cache according to the accepted tokens.

If the server's accepted tokens and additional token perfectly match a path in the proactively drafted tree—a complete draft alignment—SpecEdge retains that branch and allows the edge to continue drafting from this advanced position (Figure 5). This approach eliminates the need to regenerate already-drafted tokens, generating a deeper draft for the next drafting round by efficiently reusing prior computations.

If complete alignment fails—either because the verified path diverges from the proactive tree or doesn't reach a leaf node—SpecEdge discards the proactive work and reverts to initial drafting, rebuilding the tree from the end of the verified sequence. While this scenario doesn't benefit from proactive drafting, the strategy still improves average performance by exploiting the successful alignments when they occur.

4.3 Server-side Pipeline-aware Scheduling

Unlike conventional speculative decoding, where servers perform both drafting and verification, SpecEdge dedicates server GPUs exclusively to verification. This separation allows the server to focus on batch-based verification rather than token-by-token generation, significantly improving

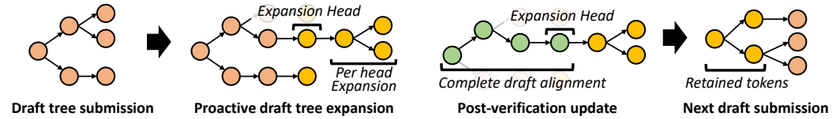


图 4：主动草稿树扩展的示例性例子。

图 5：验证后更新，包括完整的草稿对齐以及随后的草稿提交。

确保验证请求的连续流动，使昂贵的服务器 GPU 保持高度利用，而不是在等待边缘设备完成草拟时闲置。

初始草稿生成。SpecEdge 采用最先进的基于树的草稿生成方法 [Svirschevski 等人，2024]。然而，SpecEdge 被设计为面向未来，不依赖于任何特定的草稿生成技术。它仅需要一个生成候选 token 的草稿生成阶段，并且兼容各种推测解码方法，包括基于树的方法 [Chen 等人，2024, Miao 等人，2024], 有损 / 无损方法，以及早期的单序列方案 [Leviathan 等人，2023, Chen 等人，2023]。随着该领域的发展，任何推测解码的进展都可以无缝集成。

主动草稿树扩展。与在生成初始 token 后就停止的传统推测解码不同，我们的方法持续主动地进行草稿，以预测未来的服务器响应。该主动扩展的预期收益可以表示为：

$$\mathbb{E}(\text{Gain}) = P_{\text{align}} \cdot P_{\text{match}\,|\,\text{align}} \cdot \left(\frac{T_{\text{draft}}}{H_{\text{expan}}} - 1 \right) \tag{1}$$

其中 P_{align} 表示已验证序列与草稿生成序列之间对齐的概率，$P_{\text{match}\,|\,\text{align}}$ 是在对齐的情况下服务器的额外 token 与扩展头匹配的概率，T_{draft} 是主动草稿生成 token 的总数，H_{expan} 是扩展头的数量（从这些叶节点继续草稿生成）。

这种公式揭示了一个基本的权衡：增加 H_{expan} 会提高对齐概率（P_{align}），但会降低词元保留率（$\frac{T_{\text{draft}}}{H_{\text{expan}}} - 1$），而较少的头会降低对齐概率，但在对齐成功时显著提高保留率。天真地应用初始草稿生成策略 —— 将每个叶节点视为扩展头 —— 虽然最大化了 P_{align}，但收益微乎其微。我们的实证结果表明了一种违反直觉却最优的方法：在生成初始草稿树之后，SpecEdge 会识别累计对数概率最高的单一路径，并仅从该节点继续草稿生成（图 4）。这种集中策略通过在对齐发生时产生显著更高的收益来最大化预期收益，尽管对齐概率降低。

验证后更新。当服务器验证完成后，边缘 GPU 会收到已接受的 token 以及在服务器前向传播过程中生成的一个额外 token。此时，SpecEdge 会将服务器的输出与其主动生成的 token 进行比较。此外，边缘会根据已接受的 token 更新其草稿模型 KV 缓存。

如果服务器接受的 token 和额外 token 完美匹配主动草稿生成树中的一条路径 —— 一个完整的草稿对齐 ——SpecEdge 会保留该分支，并允许该边从这个高级位置继续草稿生成（图 5）。这种方法消除了重新生成已草稿生成 token 的需要，通过高效地重用先前的计算，为下一轮草稿生成生成更深入的草稿。

如果完全对齐失败 —— 无论是因为验证路径偏离了前瞻树，还是没有到达叶节点 —— SpecEdge 都会丢弃前瞻工作并恢复到初始草稿，从验证序列的末端重新构建树。虽然这种情况无法从前瞻草稿中受益，但该策略仍然通过利用成功对齐时的结果来提高平均性能。

4.3 服务器端流水线感知调度

与传统的推测解码不同，传统方法中服务器既进行草稿生成又进行验证，SpecEdge 则将服务器的 GPU 完全用于验证。这种分离使服务器能够专注于基于批次的验证，而不是逐 token 生成，从而显著提高

resource utilization. However, this disaggregated approach creates a new scheduling challenge: while the edge is busy drafting the next set of tokens, the server would idle if it only awaited verification tasks from that same request. This pattern creates "bubbles" of unutilized compute capacity whenever one part of the system is waiting on the other.

Pipeline-aware verification scheduling. SpecEdge eliminates computational inefficiencies by interleaving verification tasks across multiple requests processed on separate edge devices. The server continuously verifies completed draft batches from one set of requests while other requests simultaneously generate new drafts on their respective edge devices. This pipelined approach ensures immediate processing of incoming drafts, with verified requests promptly returning to their edge devices for additional drafting, thereby freeing server resources for the next verification batch. By aligning edge device count with server verification capacity, this orchestration effectively doubles server throughput compared to conventional server-only configurations of equivalent batch size, substantially enhancing both cost efficiency and GPU utilization.

For optimal pipeline efficiency, SpecEdge dynamically calibrates the relationship between server verification time and edge operations using real-time performance measurements. The system adjusts draft depth—the number of forward passes through the draft model—to satisfy the equation: server verification time ≈ edge drafting time + network round-trip time. This calibration ensures token batches arrive at the server precisely as it completes verifying previous batches, eliminating computational bubbles and minimizing end-to-end latency while maintaining maximum resource utilization across the distributed system.

Processing heterogeneous requests. Server-side verification must efficiently handle batches containing requests with varying sequence lengths—a common scenario when multiple users are at different stages in their generation process. SpecEdge addresses this challenge through two complementary techniques. First, it employs custom attention masking for each token sequence in the batch, ensuring the model attends only to valid tokens within that sequence while enabling parallel processing without cross-sequence interference. Second, it implements KV cache padding to match the longest sequence in the batch, avoiding the substantial computational cost of reconfiguring tensor shapes during inference despite minimal overhead for shorter sequences.

This dual approach allows SpecEdge to process diverse verification requests in unified batches, fully leveraging GPU parallelism while accommodating the asynchronous nature of edge-to-server communication. The result is maximized server throughput without sacrificing the responsiveness essential for interactive applications.

5 Evaluation

We evaluate SpecEdge in an edge-assisted server configuration against a server-only configuration across various LLMs and datasets. Our findings are summarized as follows:

- SpecEdge enhances cost efficiency by an average of **1.91×** compared to the server-only environment through increasing server throughput by **2.22×** on average.
- It reduces the inter token latency by an average of **11.24%**, even with a 14.07 ms round-trip time between the server and edge, outperforming the server-only configuration with no network delay.

Implementation and Setup. Our system's edge-assisted configuration utilizes a server-side NVIDIA A100 GPU connected to multiple edge-side NVIDIA RTX4090 GPUs over a wide-area network. The number of RTX4090 GPUs scales with the number of concurrent requests (batch size x 2). In our experiments, we measured an average round-trip time (RTT) of 14.07ms between the local edge node and our Google Cloud instance. We conducted evaluations across various models and datasets under diverse operating conditions. The code is available at `https://github.com/kaist-ina/specedge`

Baseline and metrics. Our primary baseline is a server-only configuration employing tree-based speculative decoding, supplemented by autoregressive decoding and a layer-split approach that offloads part of the LLM's layers to an edge device. SpecEdge can leverage either client-side or edge GPUs; in this evaluation, we assume it uses consumer-grade GPUs from edge cloud providers. Based on provider pricing [GCP, 2025, Vas, 2025], the server-side A100 40GB GPU costs $4.05 per hour, while running the 32B model requires an A100 80GB at $5.05 per hour. In comparison, SpecEdge adds $0.35 per hour for each RTX 4090 used. Our key metrics include cost efficiency

资源利用。然而，这种分解的方法也带来了新的调度挑战：当边缘正在草稿生成下一组 token 时，如果服务器仅等待同一请求的验证任务，它将处于空闲状态。这种模式会在系统的一部分等待另一部分时产生未利用计算能力的 “ 空档 ”。

管线感知的验证调度。SpecEdge 通过在不同边缘设备上处理的多个请求中交错执行验证任务，消除了计算效率低下的问题。服务器持续验证来自一组请求的已完成草稿批次，同时其他请求在各自的边缘设备上生成新的草稿。这种流水线方法确保了对传入草稿的即时处理，验证后的请求会迅速返回其边缘设备进行额外的草稿生成，从而释放服务器资源用于下一批验证。通过将边缘设备数量与服务器验证能力对齐，这种编排相比相同批量的传统仅服务器配置有效地将服务器吞吐量提高一倍，从而显著提升了成本效率和 GPU 利用率。

为了实现最佳的流水线效率，SpecEdge 使用实时性能测量动态校准服务器验证时间与边缘操作之间的关系。系统会调整草稿深度 —— 即通过草稿模型的前向传递次数 —— 以满足以下公式：服务器验证时间 ≈ 边缘草稿时间 + 网络往返时延。这种校准确保 token 批次在服务器完成验证前一个批次时正好到达，从而消除计算空洞，并在整个分布式系统中最大化资源利用的同时将端到端时延降至最低。

处理异构请求。服务器端验证必须高效地处理包含不同序列长度的请求批次 —— 这是当多个用户处于生成过程的不同阶段时的常见情况。SpecEdge 通过两种互补技术应对这一挑战。首先，它为批次中每个 token 序列采用自定义注意力掩码，确保模型仅关注该序列内的有效 token，同时在不干扰跨序列处理的情况下实现并行处理。其次，它实现了 KV 缓存填充，以匹配批次中最长的序列，避免在推理过程中重新配置张量形状的巨大计算开销，同时对较短序列产生的额外开销极小。

这种双重方法使 SpecEdge 能够以统一的批次处理多样化的验证请求，充分利用 GPU 并行性，同时适应边缘到服务器通信的异步特性。其结果是在不牺牲交互应用所需响应性的情况下，实现服务器吞吐量最大化。

5 评估

我们在边缘辅助服务器配置下对 SpecEdge 进行了评估，并将其与仅服务器配置在各种大语言模型和数据集上的表现进行了比较。我们的发现总结如下：

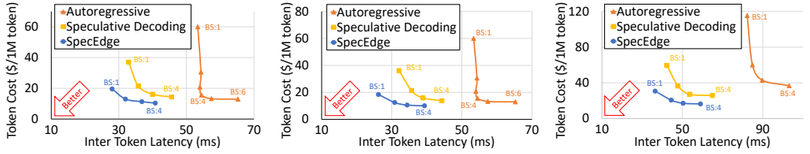
- SpecEdge 通过将服务器吞吐量提高，**与仅服务器环境相比，平均提高了 1. 效率。22×** 平均。
- 它将 token 间时延平均降低了 **11.24%**，即使在服务器与边缘之间存在 14.07ms 的往返时延，也优于没有网络时延的仅服务器配置。

实施与设置。我们系统的边缘辅助配置使用了一个服务器端的 NVIDIA A100 GPU，通过广域网连接到多个边缘端的 NVIDIA RTX4090 GPU。RTX4090 GPU 的数量会随着并发请求数（批量大小 x 2 ）而扩展。在我们的实验中，我们测得本地边缘节点与我们的 Google Cloud 实例之间的平均往返时延（ RTT ）为 14.07 毫秒。我们在不同的操作条件下，对各种模型和数据集进行了评估。代码可在 https://github.com/kaist-ina/specedge 获取。

基线和指标。我们的主要基线是仅服务器配置，采用基于树的推测解码，辅以自回归解码和分层拆分方法，将部分大语言模型的层卸载到边缘设备。SpecEdge 可以利用客户端或边缘 GPU ； 在本次评估中，我们假设它使用来自边缘云提供商的消费级 GPU。根据提供商定价 [GCP, 2025, Vas,2025]，服务器端 A100 40GB GPU 的费用为每小时 4.05 美元，而运行 32B 模型需要 A100 80GB，费用为每小时 5.05 美元。相比之下，SpecEdge 每使用一块 RTX 4090，额外增加 0.35 美元每小时。我们的关键指标包括成本效率

Table 1: Throughput and cost efficiency comparison between SpecEdge and server-only method.

Target/Draft	Task	Gen. tokens per verify		Server Throughput (tok/s)		Cost Efficiency (1k toks/$)	
		Server-only	**SpecEdge**	Server-only	**SpecEdge**	Server-only	**SpecEdge**
Qwen3 14B/1.7B	Multi-turn bench	3.92±1.51	3.98±1.57	31.78	66.54 (**2.09x**)	28.25	50.60 (**1.79x**)
	Translation	3.95±1.47	4.25±1.45	32.24	65.25 (**2.02x**)	28.66	49.47 (**1.73x**)
	Summarization	3.73±1.60	3.95±1.61	29.70	67.53 (**2.27x**)	26.40	51.22 (**1.94x**)
	QA	3.42±1.57	3.59±1.56	27.30	62.04 (**2.27x**)	24.26	47.09 (**1.94x**)
	Math.	4.10±1.48	4.25±1.49	32.84	72.93 (**2.22x**)	29.19	55.28 (**1.89x**)
	RAG	3.73±1.53	3.83±1.56	29.89	64.04 (**2.14x**)	26.57	48.78 (**1.84x**)
Qwen3 14B/0.6B	Multi-turn bench	3.87±1.41	4.41±2.25	33.45	69.58 (**2.08x**)	29.73	52.97 (**1.78x**)
	Translation	3.79±1.48	4.67±2.34	32.88	69.00 (**2.10x**)	29.22	52.23 (**1.79x**)
	Summarization	3.68±1.49	4.21±2.16	31.17	68.60 (**2.20x**)	27.71	52.16 (**1.88x**)
	QA	3.33±1.41	3.79±1.94	28.89	61.90 (**2.14x**)	25.68	46.98 (**1.83x**)
	Math	3.90±1.57	5.27±2.27	33.53	83.88 (**2.50x**)	29.80	63.56 (**2.13x**)
	RAG	3.53±1.52	4.29±2.16	30.07	69.51 (**2.31x**)	26.73	52.76 (**1.97x**)
Qwen3 32B/1.7B	Multi-turn bench	4.22±1.93	4.71±2.66	24.96	56.47 (**2.26x**)	17.80	35.38 (**1.99x**)
	Translation	4.08±1.97	5.24±2.84	24.33	58.79 (**2.42x**)	17.34	36.83 (**2.12x**)
	Summarization	4.19±2.01	4.52±2.68	24.33	54.07 (**2.42x**)	17.65	33.90 (**2.12x**)
	QA	3.62±1.95	3.93±2.49	21.59	46.14 (**2.14x**)	15.39	28.99 (**1.88x**)
	Math.	4.60±1.93	5.40±2.78	27.52	64.01 (**2.33x**)	19.62	40.18 (**2.05x**)
	RAG	3.89±2.05	4.19±2.69	22.67	49.46 (**2.18x**)	16.16	31.05 (**1.92x**)



(a) 14B-1.7B (b) 14B-0.6B (c) 32B-1.7B

Figure 6: Per token cost and inter token latency comparison between server-only baselines and SpecEdge with varying batch size (BS) and model pairs.

(generated tokens per dollar), server throughput (generated tokens per unit time) and inter token latency (user-perceived output latency). All configurations produce identical output distributions as they use the same underlying models.

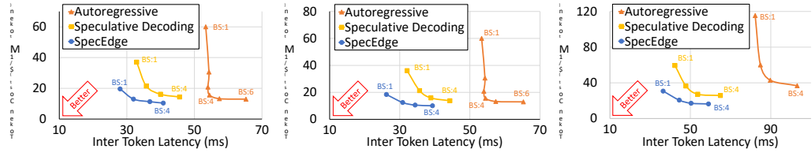
Models and data sets. We use four different LLMs: Qwen3-32B/14B [Team, 2025], Vicuna-33B [Chiang et al., 2023] and Llama2-13B-chat-hf [Touvron et al., 2023]. Unless specifically noted, all models are configured with a temperature setting of 0.7. For the draft models, we use five different models: Qwen3-1.7B/0.6B [Team, 2025], Sheared Llama-1.3B [Xia et al., 2023], Tiny Llama-1.1B [Zhang et al., 2024], and JackFram-160M [Miao et al., 2024]. Finally, we use SpecBench [Xia et al., 2024], C4 (en) [Raffel et al., 2020], OpenAssistant conversations datasets [Köpf et al., 2024], WikiText-2 [Merity et al., 2016], and MTBench [Zheng et al., 2023].

5.1 End-to-end Performance and Cost-efficiency

Table 1 presents a comparison of server-side throughput and cost efficiency between SpecEdge and a server-only speculative decoding baseline on six SpecBench tasks. We use a batch size of 1, which shows the lowest inter token latency for both SpecEdge and server-only baseline, appropriate for the latency-sensitive interactive LLM serving. Throughout the end-to-end evaluation, SpecEdge achieves **1.91×** better cost efficiency on average through **2.22×** throughput gain compared to the server-only setup. Despite the slight cost increase of employing consumer-grade edge GPUs,

Table1：SpecEdge 与仅服务器方法之间的吞吐量和成本效率比较。

Target/Draft	Task	Gen. tokens per verify		Server Throughput (tok/s)		Cost Efficiency (1k toks/$)	
		Server-only	**SpecEdge**	Server-only	**SpecEdge**	Server-only	**SpecEdge**
Qwen3 14B/1.7B	Multi-turn bench	3.92±1.51	3.98±1.57	31.78	66.54 (**2.09x**)	28.25	50.60 (**1.79x**)
	Translation	3.95±1.47	4.25±1.45	32.24	65.25 (**2.02x**)	28.66	49.47 (**1.73x**)
	Summarization	3.73±1.60	3.95±1.61	29.70	67.53 (**2.27x**)	26.40	51.22 (**1.94x**)
	QA	3.42±1.57	3.59±1.56	27.30	62.04 (**2.27x**)	24.26	47.09 (**1.94x**)
	Math.	4.10±1.48	4.25±1.49	32.84	72.93 (**2.22x**)	29.19	55.28 (**1.89x**)
	RAG	3.73±1.53	3.83±1.56	29.89	64.04 (**2.14x**)	26.57	48.78 (**1.84x**)
Qwen3 14B/0.6B	Multi-turn bench	3.87±1.41	4.41±2.25	33.45	69.58 (**2.08x**)	29.73	52.97 (**1.78x**)
	Translation	3.79±1.48	4.67±2.34	32.88	69.00 (**2.10x**)	29.22	52.23 (**1.79x**)
	Summarization	3.68±1.49	4.21±2.16	31.17	68.60 (**2.20x**)	27.71	52.16 (**1.88x**)
	QA	3.33±1.41	3.79±1.94	28.89	61.90 (**2.14x**)	25.68	46.98 (**1.83x**)
	Math	3.90±1.57	5.27±2.27	33.53	83.88 (**2.50x**)	29.80	63.56 (**2.13x**)
	RAG	3.53±1.52	4.29±2.16	30.07	69.51 (**2.31x**)	26.73	52.76 (**1.97x**)
Qwen3 32B/1.7B	Multi-turn bench	4.22±1.93	4.71±2.66	24.96	56.47 (**2.26x**)	17.80	35.38 (**1.99x**)
	Translation	4.08±1.97	5.24±2.84	24.33	58.79 (**2.42x**)	17.34	36.83 (**2.12x**)
	Summarization	4.19±2.01	4.52±2.68	24.33	54.07 (**2.42x**)	17.65	33.90 (**2.12x**)
	QA	3.62±1.95	3.93±2.49	21.59	46.14 (**2.14x**)	15.39	28.99 (**1.88x**)
	Math.	4.60±1.93	5.40±2.78	27.52	64.01 (**2.33x**)	19.62	40.18 (**2.05x**)
	RAG	3.89±2.05	4.19±2.69	22.67	49.46 (**2.18x**)	16.16	31.05 (**1.92x**)



(a) 14B-1.7B (b) 14B-0.6B (c) 32B-1.7B

图 6：不同批量大小（ BS ）和模型组合下，服务器端基线与 SpecEdge 在每个 token 成本和 token 间时延的比较。

（每美元生成的 token 数）、服务器吞吐量（每单位时间生成的 token 数）和 token 间时延（用户感知的输出时延）。所有配置产生相同的输出分布，因为它们使用相同的基础模型。

模型和数据集。我们使用四种不同的大语言模型（ LLM ）： Qwen3-32B/14B [团队，2025], Vicuna-33B [Chiang 等人，2023] 以及 Llama2-13B-chat-hf [Touvron 等人，2023]。除非特别指出，否则所有模型的温度设置为 0.7。对于草稿模型（ draft models ），我们使用五种不同的模型：Qwen3-1.7B/0.6B [团队，2025], Sheared Llama-1.3B [Xia 等人，2023], Tiny Llama-1.1B [Zhang 等人，2024], 以及 JackFram-160M [Miao 等人，2024]。最后，我们使用 SpecBench [Xia 等人，2024], C4 (en) [Raffel 等人，2020], OpenAssistant 对话 数据集 [Köpf 等 , 2024],WikiText-2 [Merity 等 , 2016], 和 MTBench [Zheng 等 , 2023].

5.1 端到端性能与成本效益

表 1 比较了 SpecEdge 与仅服务器推测解码基线在六项 SpecBench 任务上的服务器吞吐量和成本效率。我们使用批大小 1 ； 该设置在 SpecEdge 和仅服务器基线中均实现最低的 token 间时延，适合对时延敏感的交互式 LLM 服务。在端到端评测中，相比仅服务器配置，SpecEdge 通过 **2.22×** 的吞吐量提升，平均实现 **1.91×** 的成本效率提升。尽管使用消费级边缘 GPU 会略微增加成本，

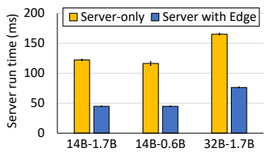


Figure 7: Server run time between server-only and server with edge drafting.

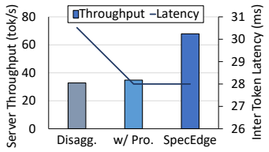


Figure 8: Performance comparison with SpecEdge components ablation.

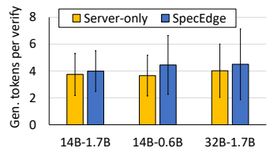


Figure 9: Generated tokens per verification between server-only and SpecEdge.

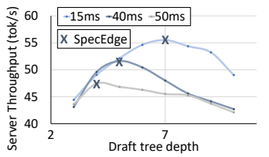


Figure 10: Server throughput according to draft depth under various network latencies.

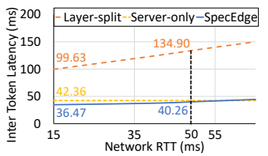


Figure 11: Inter token latency according to network round-trip time.

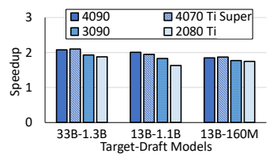


Figure 12: SpecEdge speedup gain with various edge devices upon autoregressive decoding.

SpecEdge's significant increase in server-side throughput ultimately leads to greater cost efficiency. This throughput gain is driven by server-side pipeline-aware scheduling (§4.3), which interleaves multiple requests that effectively increase server utilization, and by proactive edge drafting (§5.2), which increases the average number of generated tokens per verification. We also report the mean and standard deviation of tokens produced per verification cycle.

Beyond throughput improvements, Figure 6 demonstrates that SpecEdge also reduces inter token latency by an average of 11.24% compared to server-only speculative decoding across various batch sizes. This dual benefit—higher throughput with lower latency—is particularly valuable for interactive LLM applications where both resource efficiency and user experience are critical.

We extend our evaluation to other models, using Vicuna-33B and Llama2-13B-Chat as target LLMs across diverse datasets including C4, OAsst, WikiText-2, and MTBench. The results show that proactive edge drafting SpecEdge consistently improves performance, with greater benefits observed when the drafting models generate deeper drafts with better alignment toward the target LLMs. This alignment quality directly correlates with improvements in server throughput and reduced inter token latency. The complete results for all combinations of models and sets are provided in the Appendix B.1.

5.2 Component-wise Benefit

As illustrated in Figure 7, the server-only baseline devotes resources to both drafting and verification, causing prolonged server occupancy with each draft-verify cycle. In contrast, SpecEdge dedicates the server to verification alone while offloading drafting to more cost-effective edge devices, reducing server runtime by approximately 40-50% in all model configurations.

However, simple disaggregation alone can increase latency and leave the server underutilized (§4.1). Figure 8 compares three progressive implementations with a 14B/1.7B model pair: basic disaggregation, disaggregation with proactive edge drafting (§4.2), and complete SpecEdge with pipeline-aware scheduling (§4.3). Basic disaggregation achieves only 32.76 tokens/s throughput with higher latency, while adding proactive drafting reduces inter token latency to 28 ms. The complete SpecEdge with pipeline-aware scheduling dramatically increases server throughput to 67.89 tok/s (2.07× improvement) while maintaining latency, demonstrating the complementary benefits of each component.

Generated tokens per verification. Figure 9 compares the number of generated tokens per verification cycle between server-only speculative decoding and SpecEdge across three target–draft model pairs. On average, SpecEdge achieves 13.21% higher tokens per verification round. For the 32B–1.7B configuration, SpecEdge produces 4.5 tokens per verification compared to 4.02 with the server-only approach, while the 14B–0.6B pair sees similar gains (4.45 vs. 3.66). This efficiency gain stems from

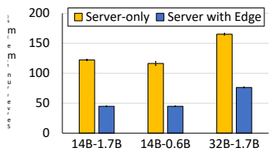


图 7：仅服务器与带边缘草稿生成服务器的运行时间比较。

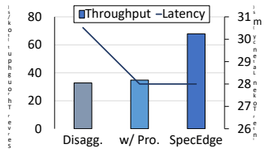


图 8：与 SpecEdge 组件消融的性能比较。

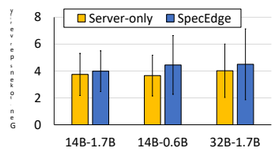


图 9：server-only 与 SpecEdge 之间每次验证生成的 token 数。

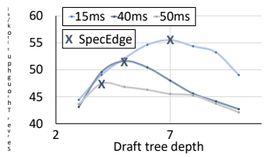


图 10：在不同网络时延下，服务器吞吐量随草稿深度的变化情况。

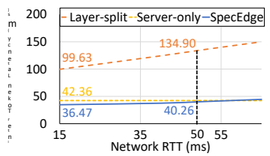


图 11：根据网络往返时延的 token 间时延。

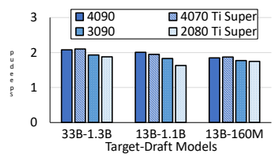


图 12：在自回归解码下，不同边缘设备上的 SpecEdge 加速增益。

SpecEdge 在服务器端吞吐量上的显著增加最终导致了更高的成本效率。这种吞吐量的提升由服务器端流水线感知调度（§4.3）驱动，该调度交错处理多个请求，有效提高了服务器利用率，同时由主动式边缘草稿生成（§5.2）驱动，增加了每次验证生成的平均 token 数量。我们还报告了每个验证周期生成的 token 的平均值和标准差。

除了吞吐量的提升外，图 6 显示，与仅服务器端推测解码相比，SpecEdge 在各种批量大小下平均将 token 间时延降低了 11.24%。这种双重收益 —— 更高的吞吐量和更低的时延 —— 对于资源效率和用户体验都至关重要的交互式大语言模型应用尤其有价值。

我们将评估扩展到其他模型，使用 Vicuna-33B 和 Llama2-13B-Chat 作为目标大语言模型（LLM），涵盖包括 C4、OAsst、WikiText-2 和 MTBench 在内的多种数据集。结果显示，主动式边缘草稿生成 SpecEdge 始终提高了性能，当草稿模型生成更深入的草稿并与目标 LLM 更好地对齐时，观察到的收益更大。这种对齐质量与服务器吞吐量的提升和 token 间时延的降低直接相关。所有模型和数据集组合的完整结果见附录 B.1。

5.2 逐组件效益

如图 7 所示，服务器 -only 基线将资源同时用于草稿生成和验证，导致每个草稿生成 - 验证周期服务器占用时间延长。相比之下，SpecEdge 将服务器专用于验证，同时将草稿生成任务卸载到成本更低的边缘设备，从而在所有模型配置中将服务器运行时间减少约 40-50%。

然而，单纯的解耦本身可能会增加时延并导致服务器未充分利用（§4.1）。图 8 对比了三种渐进式实现方法，使用 14B/1.7B 模型组合：基本解耦、带主动式边缘草稿生成的解耦（§4.2）以及带流水线感知调度的完整 SpecEdge（§4.3）。基本解耦的吞吐量仅为 32.76 tokens/s，而且时延更高，而加入主动式草稿生成后，token 间时延降低至 28ms。带流水线感知调度的完整 SpecEdge 则显著将服务器吞吐量提高到 67.89 tok/s（2.07× 提升），同时保持时延不变，展示了各组件的互补优势。

每次验证生成的 token 数量。图 9 比较了三组目标 – 草稿模型之间每个验证周期的生成 token 数量，在仅服务器推测解码和 SpecEdge 之间的差异。平均而言，SpecEdge 每轮验证生成的 token 数量高出 13.21%。对于 32B–1.7B 配置，SpecEdge 每次验证生成 4.5 个 token，而仅服务器方法生成 4.02 个 token；而 14B–0.6B 配对则看到类似的增长（4.45 对 3.66）。这种效率提升源于

proactive draft tree expansion, where each complete draft alignment allows for deeper draft trees in subsequent rounds, significantly enhancing verification efficiency and overall system performance.

Pipeline-aware draft depth adjustment. We demonstrate that dynamically adjusting the number of forward passes for edge drafting, as outlined in (§4.3), aligns the drafting phase with both server verification and network round-trip times, achieving optimal throughput in practice. Figure 10 shows how server-side throughput varies with draft tree depth under different RTTs using a 32B/1.7B model pair. On average, verification takes 94.2 ms, while each draft model forward pass needs about 11 ms. When RTT is 15 ms, SpecEdge sets the draft depth to seven; at 40 ms RTT, it sets the depth to five; and at 50 ms RTT, it decreases further to four. These results show that SpecEdge adapts draft depth for peak throughput across a range of network conditions.

5.3 System Sensitivity Analysis

Network RTT sensitivity. We evaluate the average inter token latency of SpecEdge against layer-split and server-only configurations across varying network round-trip times (RTTs), using Qwen32B on SpecBench. Layer-split configuration runs autoregressive decoding, where one-quarter of model layers run on an edge RTX4090, with the remainder on a server-side A100. The server-only configuration, unaffected by network RTT, runs tree-based speculative decoding entirely on an A100. Figure 11 shows that SpecEdge provides lower inter token latency than the server-only baseline below 50 ms RTT, with a 13.90% gain at 15 ms RTT (36.47 ms vs. 42.36 ms). Even at 65 ms, SpecEdge 's latency rises by only 22.00% over its 15 ms RTT performance (to 44.47 ms), remaining competitive. By contrast, layer-split is much slower: at 15 ms RTT, it is 2.73× slower than SpecEdge (99.63 ms vs. 36.47 ms), increasing to 3.35× slower performance at 50 ms RTT (134.90 ms vs. 40.26 ms). This resilience stems from SpecEdge 's less frequent communication rounds and proactive edge drafting, which offset network latency more effectively than layer-split approaches.

Performance with varying edge devices. Figure 12 presents the speedup achieved by SpecEdge when the server is assisted by different edge devices (RTX 4090, 4070 Ti Super, 3090, and 2080 Ti) across three target-draft model combinations. The speedup is measured relative to default autoregressive decoding using only the A100 server GPU. The consistent speedup across all model combinations confirms the architecture's robustness to different hardware configurations. As expected, more powerful edge GPUs like the RTX 4090 deliver greater speedups, while even more affordable options like the RTX 2080 Ti still provide significant acceleration. This demonstrates that SpecEdge's approach remains effective across a spectrum of edge hardware capabilities, allowing deployment flexibility. Additional results with lighter GPUs (3060 Ti, and 2080 Ti) are available in Appendix B.2.

Performance with alternative drafting approaches. We also evaluated SpecEdge with a non-tree speculative decoding approach [Leviathan et al., 2023] to demonstrate versatility beyond tree-structured methods. Using this alternative architecture, SpecEdge achieved up to 1.96× higher server throughput (Llama2 13B/TL 1.1B pairing) and 1.67× better cost efficiency compared to server-only deployments. Performance gains remained consistent across different model combinations, with even the smallest draft model (JF 160M) delivering a 1.52× throughput improvement while preserving end-user speedup. Complete results are available in Appendix B.3.

Performance with batch drafting method. We explore an alternative drafting configuration where a single edge GPU serves concurrent requests through batching. This approach enables operators to reduce the number of edge GPUs. Experiment with various batch sizes revealed a trade-off between better cost efficiency and increased latency. This configuration could be advantageous in budget-constrained deployments where latency tolerance is higher. Full results across model combinations are available in Appendix B.4.

Performance under reasoning mode. Modern LLMs provide reasoning mode for enhancing output quality, where reasoning tokens might influence speculative decoding efficiency [Wei et al., 2023, Team, 2025]. To explore SpecEdge performance under reasoning mode, we measured accepted tokens, server throughput, and cost efficiency with and without reasoning enabled. Results in Appendix B.5 show consistent improvements across all three metrics when reasoning is active, implying that our system inherently benefits from the redundancy in reasoning processes, which enhances speculative decoding performance.

Cost analysis with various GPU Providers. To ensure SpecEdge's cost-efficiency findings generalize beyond a single provider, we validated SpecEdge's performance across diverse cloud environments.

主动草稿树扩展，其中每个完整的草稿对齐都允许在后续轮次中生成更深的草稿树，从而显著提高验证效率和整体系统性能。

管线感知的草稿深度调整。我们展示了动态调整边缘草稿的前向传递次数（如 (§ 4.3) 所述），能够使草稿阶段与服务器验证及网络往返时延相匹配，从而在实践中实现最佳吞吐量。图 10 展示了在使用 32B/1.7B 模型对的情况下，不同 RTT 下服务器端吞吐量随草稿树深度的变化。平均而言，验证需要 94.2 毫秒，而每次草稿模型的前向传递大约需要 11 毫秒。当 RTT 为 15 毫秒时，SpecEdge 将草稿深度设置为七；RTT 为 40 毫秒时，将深度设置为五；而 RTT 为 50 毫秒时，则进一步降低到四。结果表明，SpecEdge 能够在不同网络条件下调整草稿深度以实现峰值吞吐量。

5.3 系统敏感性分析

网络 RTT 敏感性。我们在 SpecBench 上使用 Qwen32B，比较不同网络往返时延（ RTT ）下 SpecEdge 、按层切分配置和仅服务器配置的平均 token 间时延。按层切分配置采用自回归解码，其中四分之一模型层在边缘 RTX4090 上执行，其余模型层在服务器 A100 上执行。仅服务器配置完全在 A100 上运行树状推测解码，不受网络 RTT 影响。图 11 表明，在 RTT 低于 50 ms 时，SpecEdge 的 token 间时延低于仅服务器基线；RTT 为 15 ms 时降低 13.90% （ 36.47 ms 对比 42.36 ms ）。即使 RTT 达到 65 ms，SpecEdge 的时延也仅较 15 ms RTT 时增加 22.00% （达到 44.47 ms ），仍具有竞争力。相比之下，按层切分要慢得多：RTT 为 15 ms 时，其时延为 SpecEdge 的 2.73× （ 99.63 ms 对比 36.47 ms ）；RTT 为 50 ms 时进一步增加至 3.35× （ 134.90 ms 对比 40.26 ms ）。这种适应性来自 SpecEdge 更少的通信轮次与主动式边缘草稿生成，它们能够比按层切分方法更有效地抵消网络时延。

在不同边缘设备上的性能表现。图 12 展示了在服务器由不同边缘设备（ RTX 4090 、 4070 Ti Super 、 3090 和 2080 Ti ）协助的情况下，SpecEdge 在三种目标 - 草稿模型组合中实现的加速比。加速比是相对于仅使用 A100 服务器 GPU 进行的默认自回归解码测量的。所有模型组合中一致的加速比验证了该架构对不同硬件配置的鲁棒性。如预期，更强大的边缘 GPU 如 RTX 4090 提供了更大的加速效果，而即使是更经济的选择如 RTX 2080 Ti 仍然能提供显著的加速。这表明 SpecEdge 的方法在各种边缘硬件能力下仍然有效，允许部署具有灵活性。使用轻量级 GPU （ 3060 Ti 和 2080 Ti ）的更多结果可参见附录 B.2。

使用替代草稿生成方法的性能。 我们还评估了 SpecEdge 在非树 推测解码 方法下的表现 [Leviathan 等人， 2023] 以展示其超越树结构方法的多样性。使用这种替代架构，SpecEdge 在服务器吞吐量方面（ Llama2 13B/TL 1.1B 配对）提高了 1.96× ，在成本效率方面相比仅服务器部署提升了 1.67×。性能提升在不同模型组合中保持一致，即使是最小的 草稿模型 (JF 160M) 也实现了 1.52× 的吞吐量提升，同时保持终端用户的加速效果。完整结果请参见附录 B.3。

使用批量生成方法的性能。我们探索了一种替代的生成配置，其中单个边缘 GPU 通过批处理处理并发请求。这种方法使操作人员能够减少边缘 GPU 的数量。对各种批量大小的实验显示了更好的成本效率与时延增加之间的权衡。在预算有限且时延容忍度较高的部署中，这种配置可能具有优势。模型组合的完整结果见附录 B.4。

推理模式下的性能。 现代大语言模型提供推理模式以提升输出质量，其中推理 token 可能影响推测解码效率 [魏等 , 2023, 团队 , 2025]。为了探索 SpecEdge 在推理模式下的性能，我们测量了启用和未启用推理时的接受 token 数量、服务器吞吐量和成本效率。附录 B.5 的结果显示，当推理模式开启时，这三项指标均有一致提升，这意味着我们的系统从推理过程中的冗余中固有地受益，从而增强了推测解码性能。

使用各种 GPU 提供商的成本分析。 为了确保 SpecEdge 的成本效益发现能够超越单一提供商的适用性，我们在不同的云环境中验证了 SpecEdge 的性能。

We compared results using GPUs from multiple GPU providers (Vast.ai [Vas, 2025], Runpod [run, 2025], and TensorDock [Ten, 2025]) and Cloud Service Providers (Google Cloud Platform [GCP, 2025], Amazon Web Services [AWS, 2025], and Microsoft Azure [Azu, 2025]), accounting for the pricing variations. Across all tested configurations, SpecEdge consistently delivered cost efficiency improvements, confirming that our architectural benefits persist regardless of the specific cloud infrastructure. Detailed cross-provider comparisons are presented in Appendix C.

Detailed Case Study. Complementing our quantitative evaluation, Appendix D offers a visualization of SpecEdge in action. Using Llama models responding to a query about Dyson Spheres, we trace the complete token generation lifecycle—from initial drafting through verification and subsequent accelerated generation. The case study specifically highlights two key operational advantages: (1) how edge GPUs remain productive during server verification phases through proactive expansion, and (2) how successful draft alignments lead to deeper draft candidates in subsequent rounds.

6 Conclusion

We have presented SpecEdge, an edge-assisted LLM inference framework that leverages user-side consumer-grade GPUs for drafting candidate tokens, while the server focuses on final verification. By transmitting only finalized outputs, SpecEdge efficiently operates under typical wide-area network conditions. Proactive edge drafting on the user side maximizes the utilization of edge GPUs and reduces end-to-end latency, while pipeline-aware verification scheduling at the server ensures high throughput by efficiently aggregating and processing verification requests. Our experimental results demonstrate that SpecEdge significantly reduces operational costs by **1.91×** through delivering **2.22×** higher server throughput, and achieves a modest reduction in latency compared to server-only baselines. Overall, SpecEdge unlocks the untapped potential of powerful consumer GPUs at the edge, offering a scalable and cost-effective approach for future LLM serving deployments.

Broader Impact

This work redefines the division of labor in large language model (LLM) serving by integrating edge-side draft token generation with server-side verification, moving beyond the conventional centralized paradigm. This paradigm shift not only boosts server throughput without requiring additional data center infrastructure but also enables a novel business model for LLM services. By harnessing edge GPUs—whether through user-owned devices or edge cloud providers—our approach reduces reliance on expensive centralized servers, allowing service providers to deliver scalable, high-performance inference at a fraction of the cost. This decentralized architecture empowers businesses to adapt their infrastructure to edge resources, unlocking more flexible and cost-effective deployment strategies. Furthermore, SpecEdge alleviates the cost constraints associated with drafting, opening new avenues for speculative decoding research. By enabling the development of richer and more precise draft token generation methods, it advances the performance and capabilities of LLM services.

Limitation and Future Work

SpecEdge is designed with flexibility in mind, supporting scenarios where users may contribute their own GPUs for the edge drafting phase. Our cost-efficiency analysis focuses on deployments using consumer-grade GPUs rented from edge cloud providers, but user-owned GPUs could further enhance cost savings and scalability. Extending SpecEdge to fully leverage user-operated hardware opens up exciting opportunities for decentralized and community-driven inference. At the same time, such scenarios raise new challenges in areas such as fault tolerance and security when untrusted devices participate in computation. While SpecEdge already supports a distributed multi-user environment, exploring these broader system and security aspects is an important direction for future work.

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我们比较了使用来自多个 GPU 提供商（ Vast.ai [Vas, 2025], Runpod [run,2025], 以及 TensorDock [Ten, 2025]）和云服务提供商（ Google Cloud Platform [GCP,2025], Amazon Web Services [AWS, 2025], 以及 Microsoft Azure [Azu, 2025]）的 GPU 结果，并考虑了定价差异。在所有测试的配置中，SpecEdge 一直提供成本效益的提升，确认我们的架构优势无论在何种特定云基础设施下都能持续发挥作用。详细的跨提供商比较见附录 C。

详细案例研究。为了补充我们的量化评估，附录 D 提供了 SpecEdge 实际运行的可视化展示。通过使用 Llama 模型响应有关戴森球的查询，我们追踪了完整的 token 生成生命周期 —— 从初始草稿到验证，以及随后的加速生成。该案例研究特别突出了两个关键的操作优势：（1）在服务器验证阶段，通过主动扩展，边缘 GPU 如何保持高生产力；（2）成功的草稿对齐如何在随后的轮次中产生更深入的草稿候选。

6 结论

我们提出了 SpecEdge，一种边缘辅助的 LLM 推理框架。它利用用户侧消费级 GPU 生成候选 token，服务器则专注于最终验证。通过仅传输已生成的输出，SpecEdge 可在典型广域网条件下高效运行。用户侧的主动式边缘草稿生成充分利用边缘 GPU，并降低端到端时延；服务器端的流水线感知验证调度则通过高效聚合和处理验证请求，确保较高吞吐量。实验结果表明，SpecEdge 通过实现 **2.22×** 的服务器吞吐量提升，显著降低运营成本，获得 **1.91×** 的成本效率改善，并比仅服务器基线略微降低时延。总体而言，SpecEdge 释放了边缘侧强大消费级 GPU 尚未利用的潜力，为未来 LLM 服务部署提供了一种可扩展且经济高效的方法。

更广泛的影响

这项工作通过将边缘端的初步 token 生成与服务器端的验证相结合，重新定义了大语言模型（ LLM ）服务中的劳动分工，超越了传统的集中式范式。这一范式转变不仅在不需要额外数据中心基础设施的情况下提高了服务器吞吐量，还为 LLM 服务提供了新的商业模式。通过利用边缘 GPU—— 无论是通过用户自有设备还是边缘云提供商 —— 我们的方法减少了对昂贵集中式服务器的依赖，使服务提供商能够以极低的成本提供可扩展、高性能的推理服务。这种去中心化的架构使企业能够将基础设施适应于边缘资源，从而解锁更灵活且具有成本效益的部署策略。此外，SpecEdge 缓解了与 ··· 相关的成本限制。 草稿生成，为推测解码研究开辟新途径。通过促进更丰富、更精确的草稿 token 生成方法的发展，它推动了大语言模型服务的性能和能力。

局限性与未来工作

SpecEdge 的设计考虑了灵活性，支持用户在边缘绘图阶段贡献自己 GPU 的场景。我们的成本效益分析主要集中在使用从边缘云提供商租用的消费级 GPU 部署的情况，但用户自有 GPU 可能进一步提高成本节约和可扩展性。将 SpecEdge 扩展以充分利用用户操作的硬件，为去中心化和社区驱动的推理带来了令人兴奋的机会。与此同时，当不受信任的设备参与计算时，这类场景在容错和安全等方面也带来了新的挑战。尽管 SpecEdge 已经支持分布式多用户环境，但探索这些更广泛的系统和安全方面是未来工作的重要方向。

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A Additional Details on Evaluation Setup

For both SpecEdge and the server-only speculative decoding baseline, we adopt a tree construction algorithm from recent tree-based speculative decoding [Chen et al., 2024, Svirschevski et al., 2024]. Given a specified tree size, each forward pass of the draft model generates multiple parallel candidate tokens, which are then pruned based on cumulative log probabilities so that the total number of tokens remains within the tree budget. The total number of forward passes (the draft tree depth) is set as a hyperparameter. In our main experiments, we use the draft tree size to 32 for each request. For the server-only baseline, we find and select the optimal draft tree depth through exhaustive search, while SpecEdge determines its draft depth based on verification and network latencies as described in Section 4.3.

Throughout our evaluation, we use off-the-shelf target LLMs and draft models without any additional fine-tuning. For the main experiments, we employ Qwen3 32B/14B as the target LLM and Qwen3 1.7B/0.6B as the draft models, using the SpecBench dataset. To demonstrate broader applicability, the appendix includes experiments with Vicuna 33B and Llama2-13B-Chat-hf as target LLMs, alongside Sheared Llama 1.3B, Tiny Llama 1.1B, and JackFram 160M as draft models, evaluated on C4, OAsst, WikiText-2, and MTBench. For each query, we generate up to 256 output tokens. Since the prefill stage in SpecEdge can involve parallel processing on both server and edge, the prefill latency is determined by $\max(\text{server_time}, \text{edge_time})$ which makes metrics such as time to first token comparable to the baseline; therefore, our reported metrics focus on the output tokens after prefill. Unless otherwise specified, SpecBench (spanning six different tasks) is used throughout the experiments.

B Comprehensive Performance Analysis

B.1 Performance with Diverse Models and Datasets

In Table 2, we showcase SpecEdge 's adaptability across different target–draft model pairs and datasets, using a batch size of 1. We measure server throughput and cost efficiency gains, as well as the speedup achieved in inter token latency. Table 3 compares these speedups for both SpecEdge and a server-only speculative decoding baseline against an autoregressive decoding reference.

Our evaluation demonstrates the effectiveness of SpecEdge across several models including Qwen, Vicuna, and Llama with parameters up to 33B. While exploration with even larger models is planned as future work, we anticipate that no fundamental design changes will be needed for scaling. The

Table 2: Throughput and cost efficiency comparison between SpecEdge and server-only method.

		Gen. tokens per verify		Server Throughput (tok/s)		Cost Efficiency (1k toks/$)	
Target/Draft	Dataset	Server-only	**SpecEdge**	Server-only	**SpecEdge**	Server-only	**SpecEdge**
Vicuna 33B /SL 1.3B	C4	3.47±1.41	4.41±2.03	28.12	49.96 **(1.78x)**	24.99	38.23 **(1.53x)**
	OAsst	3.55±1.37	4.35±2.06	28.72	49.94 **(1.74x)**	25.53	37.89 **(1.48x)**
	WikiText-2	3.42±1.45	4.19±1.98	27.26	47.94 **(1.76x)**	24.23	36.42 **(1.50x)**
	MTBench	3.62±1.72	4.57±2.13	29.62	52.70 **(1.78x)**	26.33	40.21 **(1.53x)**
Llama2 13B /TL 1.1B	C4	3.40±1.35	3.48±1.34	36.80	70.76 **(1.92x)**	32.72	53.72 **(1.64x)**
	OAsst	3.57±1.40	3.67±1.39	38.55	69.49 **(1.80x)**	34.27	52.75 **(1.54x)**
	WikiText-2	3.52±1.37	3.61±1.40	37.69	70.21 **(1.86x)**	33.50	53.30 **(1.59x)**
	MTBench	3.67±1.43	3.75±1.46	40.96	76.69 **(1.87x)**	36.41	58.21 **(1.60x)**
Llama2 13B /JF 160M	C4	3.03±1.34	3.13±1.37	42.72	66.20 **(1.55x)**	37.98	50.36 **(1.33x)**
	OAsst	2.79±1.37	2.85±1.37	39.19	56.52 **(1.44x)**	34.84	42.95 **(1.23x)**
	WikiText-2	2.72±1.37	2.72±1.39	37.87	55.39 **(1.46x)**	33.66	42.08 **(1.25x)**
	MTBench	2.80±1.39	2.80±1.43	40.80	60.42 **(1.48x)**	36.42	45.92 **(1.26x)**

附加评估设置的详细信息

对于 SpecEdge 和仅服务器的推测解码基线，我们采用了来自近期基于树的推测解码的树构建算法 [Chen 等 , 2024, Svirschevski 等 , 2024]。在给定指定的树大小的情况下，每次草稿模型的前向传递会生成多个并行候选 token，然后根据累计对数概率进行剪枝，以使 token 的总数保持在树预算内。前向传递的总次数（草稿树深度）作为超参数设置。在我们的主要实验中，我们将每个请求的草稿树大小设置为 32。对于仅服务器基线，我们通过穷举搜索找到并选择最佳草稿树深度，而 SpecEdge 则根据第 4.3 节所述的验证和网络时延来确定其草稿深度。

在整个评估过程中，我们使用现成的目标大语言模型（ LLM ）和草稿模型，无需任何额外的微调。在主要实验中，我们使用 Qwen3 32B/14B 作为目标 LLM，Qwen3 1.7B/0.6B 作为草稿模型，并使用 SpecBench 数据集。为了展示更广泛的适用性，附录中包括了使用 Vicuna 33B 和 Llama2-13B-Chat-hf 作为目标 LLM，以及 Sheared Llama 1.3B 、 Tiny Llama 1.1B 和 JackFram 160M 作为草稿模型，在 C4 、 OAsst 、 WikiText-2 和 MTBench 上进行的实验。对于每个查询，我们生成最多 256 个输出 token。由于 SpecEdge 中的预推理阶段可以在服务器和边缘同时进行并行处理，因此预推理时延由 max(server_time,edge_time) 决定，这使得首 token 时延等指标与基线具有可比性；因此，我们报告的指标主要关注预推理后的输出 token。除非另有说明，实验中始终使用 SpecBench （涵盖六个不同任务）。

B 综合性能分析

B.1 在多样化模型和数据集上的性能

在表 2 中，我们展示了 SpecEdge 在不同目标 – 草稿模型对以及数据集上的适应性，使用批量大小为 1。我们测量了服务器吞吐量和成本效益提升，以及在 token 间时延上实现的加速。表 3 对 SpecEdge 和仅服务器推测解码基线相对于自回归解码参考的这些加速进行了比较。

我们的评估展示了 SpecEdge 在包括 Qwen 、 Vicuna 和 Llama 在内的多个模型上的有效性，这些模型的参数最多可达到 330 亿。虽然未来计划探索更大模型，但我们预期在扩展时无需进行根本性的设计更改。

Table2：SpecEdge 与仅服务器方法之间的吞吐量和成本效率比较。

		Gen. tokens per verify		Server Throughput (tok/s)		Cost Efficiency (1k toks/$)	
Target/Draft	Dataset	Server-only	**SpecEdge**	Server-only	**SpecEdge**	Server-only	**SpecEdge**
Vicuna 33B /SL 1.3B	C4	3.47±1.41	4.41±2.03	28.12	49.96 **(1.78x)**	24.99	38.23 **(1.53x)**
	OAsst	3.55±1.37	4.35±2.06	28.72	49.94 **(1.74x)**	25.53	37.89 **(1.48x)**
	WikiText-2	3.42±1.45	4.19±1.98	27.26	47.94 **(1.76x)**	24.23	36.42 **(1.50x)**
	MTBench	3.62±1.72	4.57±2.13	29.62	52.70 **(1.78x)**	26.33	40.21 **(1.53x)**
Llama2 13B /TL 1.1B	C4	3.40±1.35	3.48±1.34	36.80	70.76 **(1.92x)**	32.72	53.72 **(1.64x)**
	OAsst	3.57±1.40	3.67±1.39	38.55	69.49 **(1.80x)**	34.27	52.75 **(1.54x)**
	WikiText-2	3.52±1.37	3.61±1.40	37.69	70.21 **(1.86x)**	33.50	53.30 **(1.59x)**
	MTBench	3.67±1.43	3.75±1.46	40.96	76.69 **(1.87x)**	36.41	58.21 **(1.60x)**
Llama2 13B /JF 160M	C4	3.03±1.34	3.13±1.37	42.72	66.20 **(1.55x)**	37.98	50.36 **(1.33x)**
	OAsst	2.79±1.37	2.85±1.37	39.19	56.52 **(1.44x)**	34.84	42.95 **(1.23x)**
	WikiText-2	2.72±1.37	2.72±1.39	37.87	55.39 **(1.46x)**	33.66	42.08 **(1.25x)**
	MTBench	2.80±1.39	2.80±1.43	40.80	60.42 **(1.48x)**	36.42	45.92 **(1.26x)**

Table 3: Speedup of server-only, SpecEdge compared to autoregressive.

Target/Draft Model	Dataset	ITL (ms)		Speedup	
		Server-only	**SpecEdge**	Server-only	**SpecEdge**
Vicuna 33B/SL 1.3B	C4	33.229	30.612	1.86x	2.02x
	OAsst	34.496	30.460	1.76x	2.00x
	WikiText-2	34.77	30.631	1.69x	1.92x
	MTBench	33.948	31.168	1.82x	1.98x
Llama2 13B/TL 1.1B	C4	23.687	21.851	1.63x	1.76x
	OAsst	24.661	22.377	1.54x	1.70x
	WikiText-2	23.883	21.338	1.59x	1.78x
	MTBench	23.136	21.463	1.66x	1.79x
Llama2 13B/JF 160M	C4	20.6	19.007	1.87x	2.03x
	OAsst	23.711	22.378	1.60x	1.70x
	WikiText-2	26.222	24.579	1.45x	1.54x
	MTBench	23.458	22.199	1.64x	1.73x

core principles of our split computing paradigm—offloading partial decoding workloads from server to edge—naturally extend to models of any size through the draft-verify speculative decoding scheme. Our upcoming research will quantify these benefits across the full spectrum of model scales.

B.2 Performance with Lighter GPUs

We use the RTX 4090 as representative of consumer-grade GPUs, now widespread in both edge-cloud providers and user devices. However, to demonstrate broader generalizability, we conducted additional experiments with lighter consumer-grade GPUs. We measured SpecEdge performance using the RTX 3060 Ti and the RTX 2080 Ti. Table 4 shows the inter token latency and throughput with target/draft model pairs configured as Qwen3-14B/1.7B and Qwen3-14B/0.6B. Compared to the server-only approach, SpecEdge still attains meaningful throughput improvements even with less powerful GPUs.

Table 4: Performance comparison of lighter edge GPUs.

Target/Draft Model	Edge GPU	Peak FP16 TFLOPS	Memory Bandwidth (GB/s)	Inter token Latency (ms)	Server Throughput (tok/s)
Qwen3-14B/1.7B	RTX 3060 Ti	16.20	448.0	36.818	50.297
	RTX 2080 Ti	26.90	616.0	34.409	54.657
	Server-only (A100 40GB)	312	1555	32.451	30.816
Qwen3-14B/0.6B	RTX 3060 Ti	16.20	448.0	36.818	56.135
	RTX 2080 Ti	26.90	616.0	34.409	54.657
	Server-only (A100 40GB)	312	1555	32.326	30.935

B.3 Performance with Non-Tree-based Speculative Decoding Method

To demonstrate the versatility of our approach beyond tree-structured methods, we implemented the speculative decoding technique from [Leviathan et al., 2023], which uses a linear candidate sequence rather than exploring multiple branching paths. This implementation allows us to evaluate whether SpecEdge's core innovation—disaggregating drafting and verification between edge and server—generalizes effectively across different speculative decoding paradigms.

Tables 5 and 6 present the results of this evaluation, comparing SpecEdge against the conventional server-only deployment. The throughput measurements in Table 5 demonstrate SpecEdge's efficiency

表 3：仅服务器、 SpecEdge 相对于自回归的加速情况。

Target/Draft Model	Dataset	ITL (ms)		Speedup	
		Server-only	**SpecEdge**	Server-only	**SpecEdge**
Vicuna 33B/SL 1.3B	C4	33.229	30.612	1.86x	2.02x
	OAsst	34.496	30.460	1.76x	2.00x
	WikiText-2	34.77	30.631	1.69x	1.92x
	MTBench	33.948	31.168	1.82x	1.98x
Llama2 13B/TL 1.1B	C4	23.687	21.851	1.63x	1.76x
	OAsst	24.661	22.377	1.54x	1.70x
	WikiText-2	23.883	21.338	1.59x	1.78x
	MTBench	23.136	21.463	1.66x	1.79x
Llama2 13B/JF 160M	C4	20.6	19.007	1.87x	2.03x
	OAsst	23.711	22.378	1.60x	1.70x
	WikiText-2	26.222	24.579	1.45x	1.54x
	MTBench	23.458	22.199	1.64x	1.73x

我们分割计算范式的核心原则 —— 将部分解码工作从服务器卸载到边缘 —— 通过 draft-verify 推测解码方案自然地扩展到任何规模的模型。我们即将进行的研究将量化这些好处在整个模型规模范围内的影响。

B.2 轻量级 GPU 的性能

我们使用 RTX 4090 作为消费级 GPU 的代表，这类 GPU 现在在边缘云提供商和用户设备中都很普遍。然而，为了展示更广泛的可推广性，我们进行了额外的实验，使用较轻的消费级 GPU。我们使用 RTX 3060 Ti 和 RTX 2080 Ti 测量了 SpecEdge 的性能。表 4 显示了在目标 / 草稿模型对配置为 Qwen3-14B/1.7B 和 Qwen3-14B/0.6B 时的 token 间时延和吞吐量。与仅服务器的方法相比，即使使用性能较低的 GPU，SpecEdge 仍能实现显著的吞吐量提升。

表 4：轻量级边缘 GPU 的性能比较。

Target/Draft Model	Edge GPU	Peak FP16 TFLOPS	Memory Bandwidth (GB/s)	Inter token Latency (ms)	Server Throughput (tok/s)
Qwen3-14B/1.7B	RTX 3060 Ti	16.20	448.0	36.818	50.297
	RTX 2080 Ti	26.90	616.0	34.409	54.657
	Server-only (A100 40GB)	312	1555	32.451	30.816
Qwen3-14B/0.6B	RTX 3060 Ti	16.20	448.0	36.818	56.135
	RTX 2080 Ti	26.90	616.0	34.409	54.657
	Server-only (A100 40GB)	312	1555	32.326	30.935

B.3 使用非基于树的推测解码方法的性能

为验证我们的方法在树状结构以外的适用性，我们实现了 [Leviathan 等人 2023]，提出的推测解码技术。该技术使用线性候选序列，而不探索多条分支路径。这使我们能够评估 SpecEdge 的核心创新，即将草稿生成与验证分离到边缘和服务器两侧，是否能够有效推广到不同的推测解码范式。

表 5 和表 6 展示了此次评估的结果，将 SpecEdge 与传统的仅服务器部署进行了比较。表 5 中的吞吐量测量结果展示了 SpecEdge 的效率

Table 5: Server throughput and cost analysis on SpecEdge with non-tree based speculative decoding method.

	Server Throughput (tokens/s)		Cost Efficiency (1k tokens/$)	
Target/Draft Model	Server-only	**SpecEdge**	Server-only	**SpecEdge**
Vicuna 33B/SL 1.3B	24.527	39.370 (1.61x)	17.484	24.649 (1.41x)
Llama2 13B/TL 1.1B	33.044	64.851 (1.96x)	29.372	49.150 (1.67x)
Llama2 13B/JF 160M	32.755	49.639 (1.52x)	29.116	37.621 (1.29x)

Table 6: Inter token latency (ITL) comparison on SpecEdge with non-tree based speculative decoding method.

	ITL (ms)		Speedup	
Target/Draft Model	Server-only	**SpecEdge**	Server-only	**SpecEdge**
Vicuna 33B/SL 1.3B	40.771	38.067	1.61x	1.72x
Llama2 13B/TL 1.1B	30.263	26.085	1.40x	1.62x
Llama2 13B/JF 160M	30.530	30.818	1.38x	1.37x

advantages, while Table 6 quantifies the relative speedup over server-only autoregressive decoding. These results confirm that our disaggregated architecture delivers consistent improvements regardless of the underlying speculative decoding strategy, reinforcing the broad applicability of our approach.

B.4 Performance with Batch Drafting Method

While our main deployment architecture assumes that each concurrent request utilizes a dedicated edge GPU, we investigate an alternative architecture where a single edge GPU generates draft tokens for multiple concurrent requests. This alternative method offers trade-off for scenarios where operators want to utilize fewer consumer-grade edge GPUs. We evaluate this configuration by batching multiple requests on the RTX 4090 drafter with batch sizes from 2 to 4, measuring both inter token latency and cost efficiency across different target/draft model pairs.

The results show a trade-off between cost efficiency and latency. As shown in Table 7, cost efficiency improves with the alternative method, ranging from 4.4% to 29.5% better across different configurations due to better edge GPU utilization. However, this comes at the expense of increased inter token latency (5.9% to 19.0% slower), primarily caused by contention from batching multiple requests and longer draft-to-verify cycles. This alternative deployment method could be preferable in cost-sensitive scenarios where higher latency is acceptable.

Table 7: Performance comparison of alternative deployment method.

Target/Draft Model	Batch Size	Inter Token Latency (ms)	Cost Efficiency (1k toks/$)
Qwen3-14B/1.7B	2	31.777 (6.8% slower)	67.299 (14.7% better)
	3	36.617 (13.0% slower)	71.369 (29.5% better)
	4	40.099 (16.7% slower)	85.662 (24.5% better)
Qwen3-14B/0.6B	2	30.494 (8.5% slower)	67.680 (5.8% better)
	3	34.028 (15.2% slower)	80.252 (12.0% better)
	4	39.035 (19.0% slower)	82.376 (20.0% better)
Qwen3-32B/1.7B	2	43.539 (5.9% slower)	51.245 (4.4% better)
	3	48.889 (3.9% slower)	59.635 (21.3% better)
	4	58.402 (9.5% slower)	57.857 (25.1% better)

表 5：在 SpecEdge 上使用非树基推测解码方法的服务器吞吐量和成本分析。

	Server Throughput (tokens/s)		Cost Efficiency (1k tokens/$)	
Target/Draft Model	Server-only	**SpecEdge**	Server-only	**SpecEdge**
Vicuna 33B/SL 1.3B	24.527	39.370 (1.61x)	17.484	24.649 (1.41x)
Llama2 13B/TL 1.1B	33.044	64.851 (1.96x)	29.372	49.150 (1.67x)
Llama2 13B/JF 160M	32.755	49.639 (1.52x)	29.116	37.621 (1.29x)

表 6：使用非基于树的推测解码方法在 SpecEdge 上的 token 间时延 (ITL) 比较。

	ITL (ms)		Speedup	
Target/Draft Model	Server-only	**SpecEdge**	Server-only	**SpecEdge**
Vicuna 33B/SL 1.3B	40.771	38.067	1.61x	1.72x
Llama2 13B/TL 1.1B	30.263	26.085	1.40x	1.62x
Llama2 13B/JF 160M	30.530	30.818	1.38x	1.37x

优势，而表 6 量化了相对于仅服务器自回归解码的相对加速。这些结果确认了我们的分离架构无论底层推测解码策略如何，都能提供一致的改进，强化了我们方法的广泛适用性。

B.4 批处理草稿方法的性能

虽然我们的主要部署架构假设每个并发请求使用专用的边缘 GPU，但我们研究了一种替代架构，其中单个边缘 GPU 为多个并发请求生成草稿 token。这种替代方法为希望使用更少消费级边缘 GPU 的场景提供了折衷。我们通过在 RTX 4090 草稿机上对多个请求进行批处理，批量大小从 2 到 4，评估这种配置，同时测量不同目标 / 草稿模型对的 token 间时延和成本效率。

结果显示了成本效率与时延之间的权衡。如表 7 所示，由于更好的边缘 GPU 利用率，使用该替代方法的成本效率有所提高，在不同配置下提升幅度为 4.4% 到 29.5%。然而，这会带来 token 间时延增加的代价（慢 5.9% 到 19.0%），主要是由于批量处理多个请求的竞争以及较长的草稿到验证周期所致。在对成本敏感且可以接受更高时延的场景中，这种替代部署方法可能更可取。

表 7：替代部署方法的性能比较。

Target/Draft Model	Batch Size	Inter Token Latency (ms)	Cost Efficiency (1k toks/$)
Qwen3-14B/1.7B	2	31.777 (6.8% slower)	67.299 (14.7% better)
	3	36.617 (13.0% slower)	71.369 (29.5% better)
	4	40.099 (16.7% slower)	85.662 (24.5% better)
Qwen3-14B/0.6B	2	30.494 (8.5% slower)	67.680 (5.8% better)
	3	34.028 (15.2% slower)	80.252 (12.0% better)
	4	39.035 (19.0% slower)	82.376 (20.0% better)
Qwen3-32B/1.7B	2	43.539 (5.9% slower)	51.245 (4.4% better)
	3	48.889 (3.9% slower)	59.635 (21.3% better)
	4	58.402 (9.5% slower)	57.857 (25.1% better)

B.5 Performance under Reasoning Mode

Many modern LLMs support reasoning capabilities to enhance their inference performance. Reasoning tokens often exhibit highly repetitive patterns, which can impact the acceptance rate in speculative decoding. To investigate the inference performance with reasoning capabilities, we measured the accepted tokens, server throughput, and cost efficiency with and without reasoning mode enabled.

Table 8 presents the accepted tokens, server throughput, and cost efficiency for each target-draft model pair. When generating reasoning tokens, we observe consistent improvements across all three metrics: accepted tokens, server throughput, and cost efficiency. These results indicate that SpecEdge can inherently benefit from improved speculative decoding performance coming from redundancy in reasoning processes.

Table 8: Performance comparison of SpecEdge with and without reasoning mode.

	Accepted Tokens		Server Throughput (tok/s)		Cost Efficiency (1k toks/$)	
Target/Draft Model	Non-Reasoning	Reasoning	Non-Reasoning	Reasoning	Non-Reasoning	Reasoning
Qwen3-14B/1.7B	3.80 ± 1.54	4.17 ± 1.39	64.343	72.249	48.883	54.898
Qwen3-14B/0.6B	3.68 ± 1.50	4.10 ± 1.36	70.703	81.890	53.693	62.157
Qwen3-32B/1.7B	4.09 ± 1.95	4.62 ± 1.90	24.880	27.617	39.430	43.373

C Cost Efficiency with Various GPU Providers and Cloud Service Providers

We anchored our cost estimates to widely available public pricing: A100 GPUs from Google Cloud Platform and RTX 4090 GPUs from Vast.ai [Vas, 2025]. To validate the robustness of our cost-efficiency claims, we conducted additional experiments across multiple cloud environments, including various GPU providers (Vast.ai, Runpod [run, 2025], and TensorDock [Ten, 2025]) and major cloud service providers (Google Cloud Platform [GCP, 2025], Amazon Web Services [AWS, 2025], and Microsoft Azure [Azu, 2025]). Table 9 and 10 show the GPU pricing from different providers. Table 11 demonstrates that SpecEdge maintains cost efficiency improvements consistently across all tested configurations.

Table 9: Edge GPU Pricing.

GPU	GPU Provider	Cost ($/hr)
RTX 4090	RunPod	0.69
	Vast.ai	0.35
	TensorDock	0.359
RTX Pro 6000	Vast.ai	1.08
	TensorDock	1.15

Table 10: Server GPU Pricing.

GPU	Cloud Service Provider	Cost ($/hr)
A100 40GB	Google Cloud Platform	4.05
	Amazon Web Services	4.10
A100 80GB	Google Cloud Platform	5.05
	Amazon Web Services	5.12
	Microsoft Azure	3.673

D Case Study: Proactive Edge Drafting in Action

This section illustrates SpecEdge's proactive drafting mechanism through an illustrative example with a sample query. We demonstrate how the system handles the complete drafting lifecycle: initial

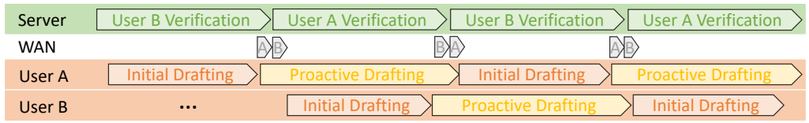


Figure 13: Timeline view showing parallel operations at edge and server, with proactive drafting occurring during server verification.

B.5 推理模式下的性能

许多现代大语言模型（ LLM ）支持推理能力，以提高其推断性能。推理 token 通常表现出高度重复的模式，这可能影响推测解码中的接受率。为了研究具有推理能力的推断性能，我们测量了启用和未启用推理模式时的接受 token 数量、服务器吞吐量和成本效率。

表 8 展示了每个目标 - 草稿模型对的接受 token 、服务器吞吐量和成本效率。在生成推理 token 时，我们观察到所有三个指标都有一致的提升：接受 token 、服务器吞吐量和成本效率。这些结果表明，SpecEdge 可以从推理过程中冗余带来的改进的推测解码性能中本质上受益。

表 8：SpecEdge 开启与关闭推理模式的性能比较。

	Accepted Tokens		Server Throughput (tok/s)		Cost Efficiency (1k toks/$)	
Target/Draft Model	Non-Reasoning	Reasoning	Non-Reasoning	Reasoning	Non-Reasoning	Reasoning
Qwen3-14B/1.7B	3.80 ± 1.54	4.17 ± 1.39	64.343	72.249	48.883	54.898
Qwen3-14B/0.6B	3.68 ± 1.50	4.10 ± 1.36	70.703	81.890	53.693	62.157
Qwen3-32B/1.7B	4.09 ± 1.95	4.62 ± 1.90	24.880	27.617	39.430	43.373

C 使用各种 GPU 提供商和云服务提供商的成本效益

我们将成本估算锚定在广泛可用的公开定价上：来自 Google Cloud Platform 的 A100 GPU 和来自 Vast.ai 的 RTX 4090 GPU [Vas, 2025]。为了验证我们关于成本效率的主张的稳健性，我们在多个云环境中进行了额外实验，包括各种 GPU 提供商（ Vast.ai, Runpod [run, 2025]，以及 TensorDock [Ten, 2025]）和主要云服务提供商（ Google Cloud Platform [GCP, 2025], Amazon Web Services [AWS, 2025]，以及 Microsoft Azure [Azu, 2025]）。表 9 和表 10 展示了不同提供商的 GPU 定价。表 11 显示 SpecEdge 在所有测试配置中始终保持成本效率的提升。

表 9：边缘 GPU 定价。

GPU	GPU Provider	Cost ($/hr)
RTX 4090	RunPod	0.69
	Vast.ai	0.35
	TensorDock	0.359
RTX Pro 6000	Vast.ai	1.08
	TensorDock	1.15

表 10：服务器 GPU 定价。

GPU	Cloud Service Provider	Cost ($/hr)
A100 40GB	Google Cloud Platform	4.05
	Amazon Web Services	4.10
A100 80GB	Google Cloud Platform	5.05
	Amazon Web Services	5.12
	Microsoft Azure	3.673

D 案例研究：主动式边缘草稿生成实践

本节通过带有示例查询的说明性示例说明了 SpecEdge 的主动草稿生成机制。我们演示了系统如何处理完整的草稿生成生命周期：初始

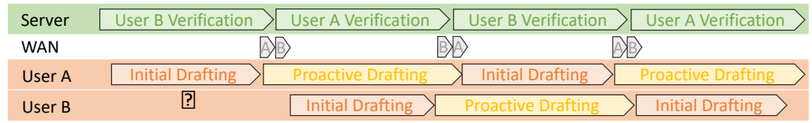


图 13：时间轴视图显示了边缘和服务器的并行操作，同时在服务器验证期间发生主动草稿生成。

Table 11: Cost efficiency comparison of GPU Providers and Cloud Service Providers.

Target/Draft Model	Server/Edge GPU	GPU Provider	Cost Efficiency (1k toks/$)		
			Google Cloud Platform	Amazon Web Service	Microsoft Azure
Qwen3-14B/1.7B	A100 40GB / RTX 4090	Vast.ai	51.018 (1.87×)	50.485 (1.87×)	-
		RunPod	44.721 (1.64×)	44.311 (1.65×)	-
		TensorDock	50.829 (1.87×)	50.30(1.87×)	-
		Baseline (Server-only)	27.222	26.890	-
Qwen3-14B/1.7B	A100 40GB / RTX Pro 6000	Vast.ai	47.68 (1.75×)	47.30 (1.76×)	-
		TensorDock	46.64 (1.71×)	46.27 (1.72×)	-
		Baseline (Server-only)	27.222	26.890	-
Qwen3-14B/0.6B	A100 40GB / RTX 4090	Vast.ai	52.938 (1.86×)	52.386 (1.88×)	-
		RunPod	46.382 (1.63×)	45.957 (1.65×)	-
		TensorDock	52.741 (1.86×)	52.192 (1.88×)	-
		Baseline (Server-only)	28.415	27.802	-
Qwen3-32B/1.7B	A100 80GB / RTX 4090	Vast.ai	31.619 (1.82×)	31.332 (1.83×)	41.757 (1.75×)
		RunPod	28.453 (1.64×)	28.144 (1.65×)	36.280 (1.52×)
		TensorDock	31.619 (1.82×)	31.239 (1.83×)	41.591 (1.75×)
		Baseline (Server-only)	17.330	17.090	23.822

draft generation, proactive expansion, server verification, and subsequent drafting. Figure 13 shows the timeline view of each operation.

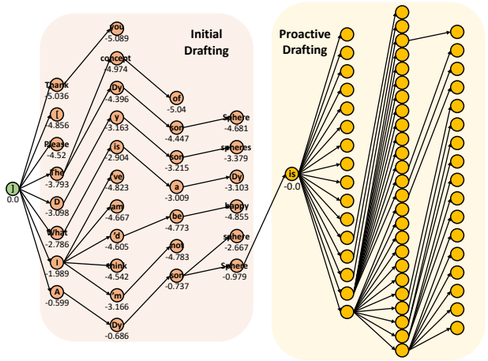


Figure 14: Example of proactive draft tree expansion after initial drafting. After creating the initial draft tree (left), the edge device identifies the most probable leaf token and proactively generates additional tokens (right) while awaiting server verification.

Experimental Setup. For this demonstration, we use a sample from the OAsst dataset with the query "What is a Dyson Sphere?". We employ Llama-3.2-3B [Grattafiori et al., 2024] as the target model and Llama-3.2-1B as the draft model.

Initial Draft and Proactive Expansion. Figure 14 illustrates the transition from initial drafting to proactive expansion. Once the edge device constructs the initial draft tree, it submits this tree to the server for verification. Simultaneously—rather than remaining idle—the edge identifies the expansion head with the highest cumulative log probability and begins generating additional tokens proactively.

Server Verification and Alignment. When the verification results arrive from the server, the edge device compares its locally expanded tree with these results. Figure 15 demonstrates a case of complete draft alignment, where the server's verified tokens ("A Dyson Sphere is") match both the initial draft tree path and the selected expansion head.

表 11：GPU 提供商与云服务提供商的成本效率比较。

Target/Draft Model	Server/Edge GPU	GPU Provider	Cost Efficiency (1k toks/$)		
			Google Cloud Platform	Amazon Web Service	Microsoft Azure
Qwen3-14B/1.7B	A100 40GB / RTX 4090	Vast.ai	51.018 (1.87×)	50.485 (1.87×)	-
		RunPod	44.721 (1.64×)	44.311 (1.65×)	-
		TensorDock	50.829 (1.87×)	50.30(1.87×)	-
		Baseline (Server-only)	27.222	26.890	-
Qwen3-14B/1.7B	A100 40GB / RTX Pro 6000	Vast.ai	47.68 (1.75×)	47.30 (1.76×)	-
		TensorDock	46.64 (1.71×)	46.27 (1.72×)	-
		Baseline (Server-only)	27.222	26.890	-
Qwen3-14B/0.6B	A100 40GB / RTX 4090	Vast.ai	52.938 (1.86×)	52.386 (1.88×)	-
		RunPod	46.382 (1.63×)	45.957 (1.65×)	-
		TensorDock	52.741 (1.86×)	52.192 (1.88×)	-
		Baseline (Server-only)	28.415	27.802	-
Qwen3-32B/1.7B	A100 80GB / RTX 4090	Vast.ai	31.619 (1.82×)	31.332 (1.83×)	41.757 (1.75×)
		RunPod	28.453 (1.64×)	28.144 (1.65×)	36.280 (1.52×)
		TensorDock	31.619 (1.82×)	31.239 (1.83×)	41.591 (1.75×)
		Baseline (Server-only)	17.330	17.090	23.822

草稿生成、主动扩展、服务器验证以及随后的草稿生成。图 13 显示了每个操作的时间线视图。

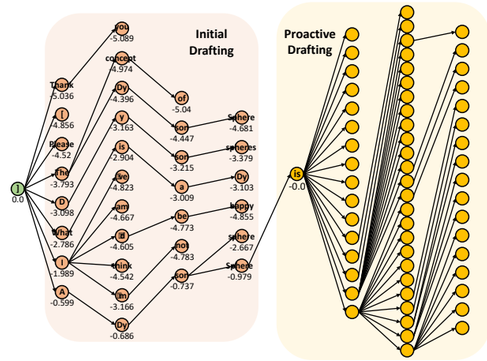


图 14：初始草稿生成后的主动草稿树扩展示例。在创建初始草稿树（左）后，边缘设备识别最可能的叶子 token，并在等待服务器验证的同时主动生成额外的 token （右）。

实验设置。在本演示中，我们使用来自 OAsst 数据集的一个样本，查询为 “ 戴森球是什么？ ”。我们使用 Llama-3.2-3B [Grattafiori 等人，2024] 作为目标模型，并使用 Llama-3.2-1B 作为草稿模型。

初稿与主动扩展。图 14 展示了从初稿草稿生成到主动扩展的过渡。一旦边缘设备构建了初稿树，它会将该树提交给服务器进行验证。与此同时 —— 而不是保持空闲 —— 边缘设备会识别具有最高累计对数概率的扩展节点，并开始主动生成额外的 token。

服务器验证与对齐。当来自服务器的验证结果到达时，边缘设备将其本地扩展的树与这些结果进行比较。图 15 展示了一个完整草稿对齐的案例，其中服务器验证的 token （ "A Dyson Sphere is" ）与初始草稿树路径及所选扩展节点均匹配。

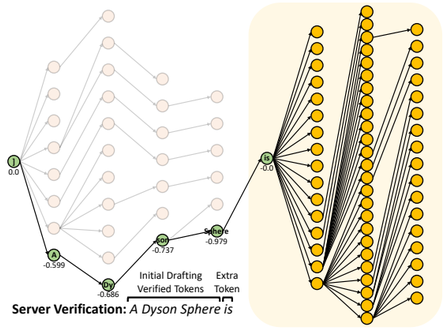


Figure 15: Server verification and complete draft alignment. The server verification results (showing "A Dyson Sphere is") perfectly align with a path in the draft tree, validating both the initial draft and the chosen expansion head.

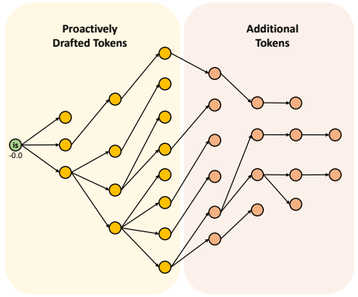


Figure 16: Deeper token drafting with proactively drafted tokens. The previously generated proactive tokens tree following the extra token ("is") can be immediately used in the next drafting cycle without additional computation.

Accelerated Token Drafting. Following successful verification, SpecEdge leverages the proactively drafted tokens to benefit the next drafting cycle. Figure 16 shows how these pre-generated tokens contribute to the next draft submission, eliminating the need to regenerate these tokens and thus building deeper draft candidates.

Performance Implications. This example demonstrates how SpecEdge's proactive drafting strategy provides tangible performance benefits:

- **Reduced idle time:** The edge GPU remains productive during server verification periods.
- **Deeper subsequent drafting:** Pre-generated tokens allow additional draft forward passes for the next drafting cycle.
- **Effective resource utilization:** Computational resources on both edge and server are maximized.

The complete alignment case shown here represents the optimal scenario, though SpecEdge handles partial or misaligned drafts as well (Section 4.2).

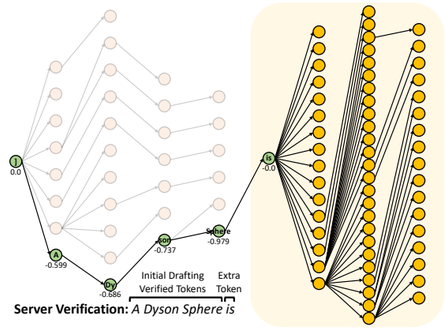


图 15：服务器验证和完整草稿对齐。服务器验证结果（显示“一个戴森球是”）与草稿树中的路径完美对齐，验证了初始草稿和所选择的扩展节点。

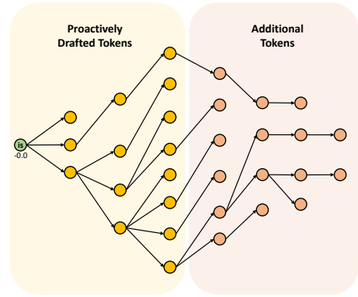


图 16：通过主动生成的 token 进行更深入的 token 草稿生成。之前生成的沿着额外 token（“is”）的主动 token 树可以在下一轮草稿生成中立即使用，而无需额外计算。

加速 token 草稿生成。在成功验证之后，SpecEdge 利用主动草稿生成的 token 来提高下一个草稿生成周期的效率。图 16 显示了这些预生成的 token 如何有助于下一次草稿提交，从而无需重新生成这些 token，并因此构建更深的草稿候选项。

性能影响。此示例演示了 SpecEdge 的前瞻性草稿生成策略如何提供切实的性能收益：

- **减少空闲时间：**在服务器验证期间，边缘 GPU 仍保持高效工作。
- **更深入的后续草稿生成：**预生成的 token 允许在下一个草稿生成周期中进行额外的草稿前传。
- **有效的资源利用：**边缘和服务器上的计算资源都得到了最大化利用。

这里显示的完全对齐案例代表了最优情形，尽管 SpecEdge 同样处理部分或未对齐的草稿（第 4.2 节）。

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   回答：[是]

   理由：我们指定每个实验设置使用的计算资源类型。

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   - 答案 NA 意味着论文中没有包括实验。
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